Qualia Motion in Fourier Space: Formalizing Linear, Nondirected and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1

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In this article, we formalize and analyze qualia motion, i.e., the process by which a composition transitions across distinct harmonic qualities through the *Fourier qualia space* (FQS)—a multidimensional and transposition-independent space based on the discrete Fourier transform (DFT) coefficients' magnitude. In the FQS, the plot of set classes relies on their harmonic qualities—such as diatonicity and octatonicity—enabling us to (1) identify the pitch-class set in a musical phrase that best represents its qualia—a reference sonority; (2) define a harmonic progression using all sequential reference sonorities in a piece, (3) visualize trajectory in space, and (4) establish a statistical metric for the ambiguity of harmonic qualia. Finally, we discuss Schoenberg's Op. 19, No. 1, analyzing the sense of its harmonic path. The proposed space leverages a bipartite, symmetrical, and consequential structure and unveils ambiguity as an element of nondirected linearity and counterpoint.

Keywords: harmonic qualities, ambiguity, harmony, linearity, discrete Fourier transform, Schoenberg, free atonality, harmonic spaces, reference sonority.

INTRODUCTION

Ian Quinn's concept of *chord quality* refers to the abstract relationship emerging between chords (set classes) and chord qualia[1] as projected in *quality space*, i.e., a spatial model for chord classification under similarity relations.[2] Such relationships classify pitch-class sets (pcsets) as more or less similar to six referential harmonic qualities.[3] Several analytical studies have used the framework of harmonic qualia to address the complex harmonic space of Schoenberg's free atonal period.[4] For instance, Jack Boss and Joshua Mailman

We would like to thank the article's reviewers whose critical considerations helped us shape our manuscript. We also thank Elisama, Rodrigo, and Eduardo for their meticulous review and valuable suggestions. Lastly, we acknowledge the University of Porto (UP), the Faculty of Engineering (FEUP), and INESC TEC for providing us with the necessary facilities and health support that enabled us to conduct our work. The Foundation for Science and Technology (FCT) provided research funding for this project under doctoral scholarship No. 2021.05059.BD.

[1]In the following pages, we employ the term qualia (or quale) to refer to harmonic qualities such as diatonicity or octatonicity (Harding [2021, 7]), diverging from its conventional usage referring to subjective consciousness experiences. Despite the subjective connotations of qualia/quale, we use the DFT for its quantification, thus systematizing a method to assess chord quality.

[2]For a detailed description of the *chord quality* concept and the *quality space*, please refer to Quinn (2006, 2007).

[3]Several authors discuss how the magnitude of the six non-trivial DFT coefficients resonate with six distinct harmonic qualities, including Quinn (2007), Yust (2016), Chiu (2021), and Harding (2021).

[4]Although Schoenberg avoided the term "atonal," arguing that it would describe music lacking any tones, others applied it to describe the music of his second and third compositional periods (Schoenberg [1983, 432–33]; Simms [2000, 8–10]; Boss [2019, 1]).

unpacked the intricate vertical web of Schoenberg's free atonal pieces by studying their harmonic qualia.[5] In fact, these authors' analyses contain a unifying factor: the inference of progressions of harmonic qualities that express and represent a macro perspective of the pieces' harmonic unfolding.[6]

As such, the possibility to formalize qualia motion in an algebraic or geometric space offers several advantages to analytical practice.[7] The most significant advantage is the ability to measure metaphorical and subjective analytical concepts, such as the distance between pcsets.[8] In recent years, the discrete Fourier transform (DFT) of pcsets has become established as a mathematical-theoretical framework for inferring musical style or musical entropy.[9] More relevant to the work we present here is the DFT's potential to (1) enable the development of analytical spaces for both pitch and set classes and (2) establish a relationship between the DFT coefficients' magnitude and the six harmonic qualities—chromaticity, dyadicity, triadicity, octatonicity, diatonicity, and whole-tone quality.[10] Note that while DFT phase spaces are sensitive to transpositions, DFT magnitude spaces (based on DFT coefficients' magnitudes) are instead sensitive to variations in the intervallic

[5]Concerning Schoenberg's Op. 19, Boss (2019, 224–62) presents analyses of Nos. 2, 3, and 6 and Mailman (2015) of No. 3.

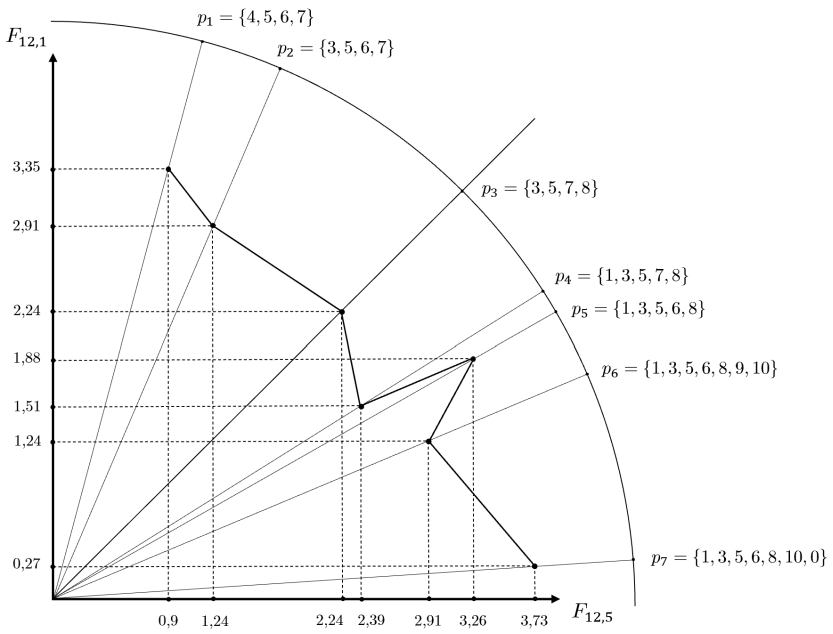
[6]A different perspective on the study of harmonic qualities is Martins's (2023) work on Witold Lutosławski's mid-century harmony, examining the relation between chord color and harmonic motion. The study characterizes twelve-tone chordal space as the interaction between interval patterns, registral deployments and temporal orderings, examining how these interactions imprint distinct coloristic qualities to single chordal entities, while also being implicated in strategies of harmonic progression, relatedness, complementarity, and contrast.

[7]Hook (2023, 4) defines *space* as "a collection of musically meaningful entities in some visual or geometric arrangement." To clarify, we employ *space* to specifically refer to its mathematical and geometric facets, typically comprised of n spatial dimensions, where different mathematical objects are points of n dimensions.

[8]Hook (2023, 3–4) and Tymoczko (2011, 65) argue that geometric spaces can offer multiple advantages to musical analysis by bringing abstract musical concepts to a more accessible plane through visual representation (a point Hook notes is particularly important given humans' predominant visual orientation). For example, geometric spaces can clarify complex musical principles like voice-leading and contrapuntal relations, whilst providing tools that map temporal and pitch-based phenomena onto spatial dimensions. Moreover, Yust (2015b, 121), introducing his DFT-based *phase space*, states that a harmonic space not only allows the visualization of harmonic objects but also permits the assessment of analytical ideas such as distance, regions, boundaries, and trajectories.

[9]Numerous studies showcase the versatility of the DFT in music-related analyses. For an overview of its application in pcset analysis, voice-leading, homometry, and rhythm, consult Amiot (2017a). Additionally, Amiot (2021) proffers a pioneering framework utilizing the DFT for assessing pcset entropy. Further insights into harnessing DFT-based audio features for effective classification of distinct musical styles can be found in Almeida, Bernardes, and Weiß (2022). Bernardes et al. (2016) elucidate a DFT space wherein Euclidean distances from the center indicate consonance, while Bernardes, Carvalho, and Pereira (2023) present a DFT-based harmonic lexicon in the *FluidHarmony* assisted-composition tool. In a related study, Callender (2007) employs pcsets' DFT spectra to situate chords within a harmonic space, thereby facilitating the definition of harmonic similarity measures for any tuning system. Lastly, Dawson, Perez, and Sylvestre (2020) reveal a striking correlation between DFT phase spaces and artificial neural networks trained in musical tasks, despite the absence of an explicit mathematical connection.

[10]Several authors discuss the association between the non-trivial DFT coefficients and six harmonic qualities, namely Bernardes, Carvalho, and Pereira (2023), Chiu (2021), Harding (2021), Laneve et al. (2023), Quinn (2007), Yust (2015b, 2016, 2019, 2020), and Yust, Lee, and Pinsky (2022).



EXAMPLE 1. *Illustration exemplifying Quinn's (2007) quality space, where the x and y-axes represent the DFT coefficients 5 and 1, respectively. The author models the transition from chromaticity to diatonicity in Ligeti's Lux Aeterna with seven chords from the piece's opening.*

structure of the pcset under analysis.[11] As a result, pcsets belonging to a particular set class are coincident in DFT magnitude spaces, regardless of their pitch classes.[12]

Quinn's work provides the inspiration and theoretical foundations for the following pages.[13] It establishes the relationship between harmonic qualities and relevant DFT coefficients and advances strategies to formalize qualia motion in Fourier space. In his *quality space*, it is possible to define the trajectory of a work between two DFT coefficients, i.e., two harmonic qualities. Example 1 exhibits Quinn's *quality space* and the opening transition in Ligeti's *Lux Aeterna* from a chromatic to a diatonic harmonic quality. However, analyzing a work's trajectory in space using all six harmonic qualities associated with DFT coefficients is a challenging task due to the limitations of human perception within three spatial dimensions. To address this problem, Jennifer Harding opted for an analytical strategy that overlaps all six DFT coefficients' magnitude in the same graphic.[14] Usually, this representation uses distinct colors for each coefficient, allowing a clear visualization of the DFT magnitudes unfolding throughout a piece. However, beyond its difficult interpretation, this approach lacks the interaction between DFT coefficients. It blurs Quinn's idea of a *quality space* in which pcsets become harmonic objects whose distances and positioning specify their *chord quality*. As such, the possibility of reducing DFT's multidimensional nature to an intelligible two-dimensional space would allow

[11]Yust (2015a, 2018).

[12]Take, for example, the C major and D major chords. They consist of distinct pitch classes and would occupy different places in a phase space (Amiot [2013]; Yust [2015b]), but since both belong to the same set class, they are coincident in a DFT-magnitudes space.

[13]Quinn (2006, 2007).

[14]Harding (2020, 99).

(1) the expansion of Quinn's *quality space* for all six harmonic qualities associated with DFT coefficients, (2) the analysis of the *chord quality* of any pcset or set class through its spatial relations, (3) better interpretability of the space, and (4) would foster musical analyses where the qualities mutually interact and influence each other, providing a more comprehensive overview of the pieces' qualia motion.

Nevertheless, the definition of a space where the six harmonic qualities coexist can also leverage ways to calculate qualia ambiguity, that is, to understand how much a group of set classes is (or is not!) associated with a specific quale. In other words, such a space can pave the way for the disambiguation of ambiguity, bringing it to a more objective and measurable plane. *Ambiguity* refers to the principle by which something has two or more mutually exclusive ways of being perceived. Inevitably, in music, the term carries a long compositional and analytical history.[15] For example, analysts relate ambiguity with metrics regarding an unclear perception of downbeats, but also with pitch collections or tonality, meaning that the same pitches have different interpretations in distinct modes or tonalities.[16] Sometimes, composers intentionally seek ambiguity as a tool capable of producing more poetic musical discourses with a multiplicity of interpretations.[17] The ambiguity we seek to describe is a concept close to how David Temperley discusses tonal ambiguity, i.e., "the degree to which a set implies one key as opposed to several."[18] In parallel reasoning, a quality space can then foster the idea of qualia ambiguity—a measure of how much a set class is associated with a harmonic quality to the detriment of others.

The problem of reducing multidimensional data has been a focus of research in mathematics and computation in recent years.[19] As a result, researchers devised multiple techniques, including *Principal Component Analysis*, *Isomap*, *Multidimensional Scaling*, and *t-SNE*.[20] These techniques usually generate a reduced space that preserves data structure, thus leveraging an interpretation of the pcsets' relative position. However, the dimensions in the reduced space do not correspond directly to the original dimensions, making it challenging to ascertain the relationship between the reduced coordinates and the original multidimensional coordinates.

In this context, our paper proposes the following four contributions. To address the definition of a quality space where all six DFT dimensions interact, we advance (1) the *Fourier qualia space* (FQS), a two-dimensional model of the full range of harmonic qualities in DFT. Inspired by Mailman's *Representational Hierarchy*

[15]For an overview of the concept of ambiguity in multiple musical domains and its pedagogical implications see Karpinski (2012).

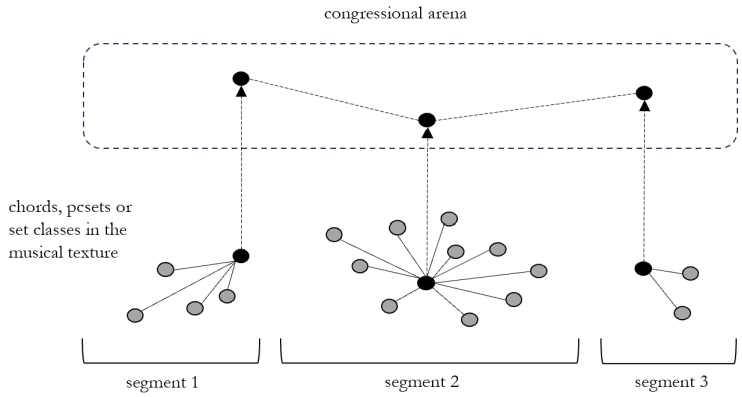
[16]Please examine Karpinski (2012) for a discussion of metric ambiguity and Temperley (2007; 2006, 116–20) for a definition of tonal ambiguity.

[17]To achieve this result, composers exploit the perceptual state of the listener. Meyer (1956, 51–52) and Thomson (1983) argue that a genuinely ambiguous musical passage necessitates not only a duality or multiplicity of musical interpretations but also a state of uncertainty in the listener's mind regarding the most accurate interpretation.

[18]Temperley (2007, 24).

[19]For an overview of various techniques for multidimensional data visualization, please refer to Dzemyda, Kurasova, and Žilinskas (2013) and Wilkinson (2005, 248–53).

[20]For Principal Component Analysis (PCA), consult Hotelling (1933) and Pearson (1901). For Isomap, see Tenenbaum, Silva, and Langford (2000). For Multidimensional Scaling (MDS), please refer to Torgerson (1952). For t-Distributed Stochastic Neighbor Embedding (t-SNE), see Hinton and Roweis (2002) and Maaten and Hinton (2008).



EXAMPLE 2. *Diagram for Mailman's (2015) Representational Hierarchy Association model, which materializes in (1) the segmentation of a piece into multiple parts; (2) the selection of a representative event for each segment; and (3) the comparison of the selected events in an associative arena or space.*

Association method, illustrated in Example 2, the FQS is the foundation of a computational method that (2) selects the most representative pcset of a phrase's harmonic qualia and is the arena where all representative pcsets are associated.[21] Also leveraged by the FQS, we (3) define a metric to determine the qualia ambiguity of a musical phrase. Finally, the previous contributions provide (4) an analytical perspective for the first piece of Schoenberg's *Sechs Kleine Klavierstücke*, Op. 19, No. 1.[22]

In detail, the FQS uses the Radial Visualization (*RadViz*) technique.[23] It projects six-dimensional data in a regular polygon with vertices that become anchors to springs. Any harmonic object projected in the FQS attaches to these springs with forces given by its DFT multidimensional coordinates. As such, the FQS defines a hexagonal quality space whose vertices map to the six relevant DFT coefficients. We also identify six regions corresponding to chromaticity, dyadicity, triadicity, octatonicity, diatonicity, and whole-tone quality. Thus, the region a given set class occupies in space defines its link to each harmonic quality. On the other hand, the intrinsic properties of the FQS allow us to advance a mathematical method inspired by Mailman's *Representational Hierarchy Association*, to calculate the pcset that best represents and summarizes all chord qualities in a musical phrase—a reference sonority (RS). This calculation is intrinsic to the position of that phrase's pcsets in FQS.

In addition, we propose a statistical metric to measure the qualia ambiguity of any collection of pcsets, taking advantage of the fact that the FQS's center stands as a point without a clear quale resulting from the balance between the six harmonic qualities in a convex space. As such, the closer a group of pcsets is to the center,

[21]Mailman (2015).

[22]The fact that the FQS is at the center of our methods echoes Quinn's (2007, 61) argument about the analytical and theoretical prolificacy of his quality space as its best justification: "What justifies this spatial model in the end is the supporting conceptual framework itself, which seems to be quite stimulating in both analytical and theoretical directions."

[23]A detailed explanation of the *RadViz* technique and its mathematical constructs can be found in Dzemyda, Kurasova, and Žilinskas (2013, 9–12), Inselberg and Dimsdale (1987), and Rubio-Sanchez et al. (2016).

the more ambiguous its qualia will be. Conversely, locations closer to the anchors have low ambiguity. Finally, we approach Schoenberg's Op. 19, No. 1 analytically with the caveat that while music summons an intricate network of relations between structural dimensions such as melody, harmony, rhythm, form, and texture, our article focuses on harmony only, as if in a *ceteris paribus* analysis. This single-topic approach has the advantage of assessing the impact of a single variable on a complex musical entity full of relationships. With these analytical ideas, we intend to understand the role of harmonic qualities in unfolding the piece's musical discourse, unveil its path in FQS, and exhibit the development of qualia ambiguity in each phrase. Moreover, our analysis establishes a dialogue between our results and Jonathan Kramer's concept of nondirected linearity.[24]

The article is organized as follows: Section 1 describes the DFT of pcsets and Mailman's methodological conceptions; Section 2 details the construction of the FQS; Section 3 unveils the computation of an RS and ambiguity; Section 4 weaves analytical considerations on Schoenberg's Op. 19, No. 1; and finally, we present the conclusions of our study and questions for future work in Section 5.

1 THE FOURIER TRANSFORM AND REPRESENTATIONAL HIERARCHY ASSOCIATION: A REVIEW

In this section, we present two theoretical elements that form the basis of our article: (1) the mathematical framework commonly known as DFT of pcsets—with a particular emphasis on the magnitude of DFT coefficients—and (2) Mailman's analytical methodology, which aims at establishing criteria to segment a piece into smaller parts, selecting a pcset as RS for each segment, and defining an abstract space where the RSs can be compared.

1.1 THE DISCRETE FOURIER TRANSFORM OF PITCH-CLASS SETS

The DFT is one of the most potent mathematical techniques in image and sound processing.[25] It allows decomposing a discrete and infinitely periodic complex signal into multiple elementary signals—DFT coefficients. When summed, they return the original signal.

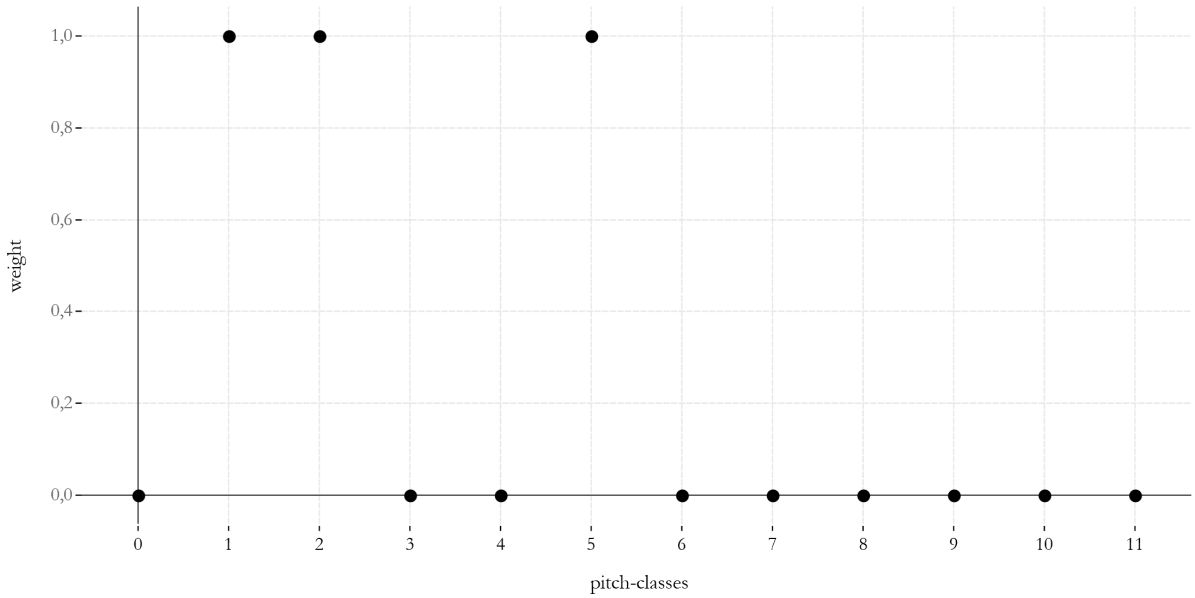
This technique also applies to pcsets, as suggested by David Lewin, in an analogous process to the one described in the previous paragraph.[26] However, unlike an audio signal, where the x-axis represents time, in the DFT of pcsets, the x-axis represents the n parts—$\mathbb{Z}_n$—into which the octave divides. In $\mathbb{Z}_{12}$, any pcset X can be written as $X = (x_0, x_1, \dots, x_{11})$, where $x_0, x_1, \dots, x_{11}$ are binary numbers denoting the presence of the twelve pitch-classes in X.[27] Example 3 shows a graphical overview of X, trichord {1,2,5}, so that $X = (0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0)$, consisting of twelve binary numbers of an infinitely periodic signal repeated at

[24]Kramer (1988).

[25]For convenience, Bracewell (2000, 258–85) provides a mathematically detailed compendium of Fourier transform applications.

[26]Lewin (1959, 2001).

[27]Note that we intentionally chose binary encoding for calculating the DFT, over counting the absolute value of the pitch-classes in a pcset. We made this choice because (1) the compositional practice of free atonality avoids duplicating pitch-classes and avoids intervals such as the unison and perfect octaves, and (2) it allows for placing the set class at the center without biasing any of its subsets.



EXAMPLE 3. *The depiction of pcset {1,2,5} is a binary distribution in which ones and zeros denote the presence or absence of the twelve pitch classes.*

each octave.[28] The DFT coefficients are complex numbers denoted as $\hat{f}_0, \hat{f}_1, \ldots, \hat{f}_{11}$, each corresponding to the decomposition of X in twelve distinct coefficients.[29] Calculating $\hat{F}(X)$ enables the reconstruction of the signal by summing up all its DFT coefficients. Such coefficients are complex numbers defined by $a + bi$ with real and imaginary parts. They also define themselves with magnitude $\left|\hat{f}_k\right|$ and phase φ_k values.[30] Hence, $\left|\hat{f}_k\right|$ and φ_k allow one to visualize any DFT coefficient as a sinusoid in a Cartesian plane, where $\left|\hat{f}_k\right|$ defines the sinusoid's amplitude, and φ_k its horizontal shift.[31] Each sinusoid can be interpreted as the best approximation of the original pcset given its period.[32] Furthermore, the number of periods per octave for each coefficient $\hat{f}_k$ is k, for $k > 0$, since $\hat{f}_0$ reflects the pcset cardinality.

Example 4(a) depicts the $\hat{F}(X)$ sinusoids $\hat{f}_3$ and $\hat{f}_4$, with three and four periods per octave. Example 4(b) shows the sum of the coefficients $\hat{f}_3$ and $\hat{f}_4$ for pcset X (trichord {1,2,5}). Even with only two DFT coefficients summed, one can see how the signal in pitch-classes 1, 2, and 5 starts to stand out.

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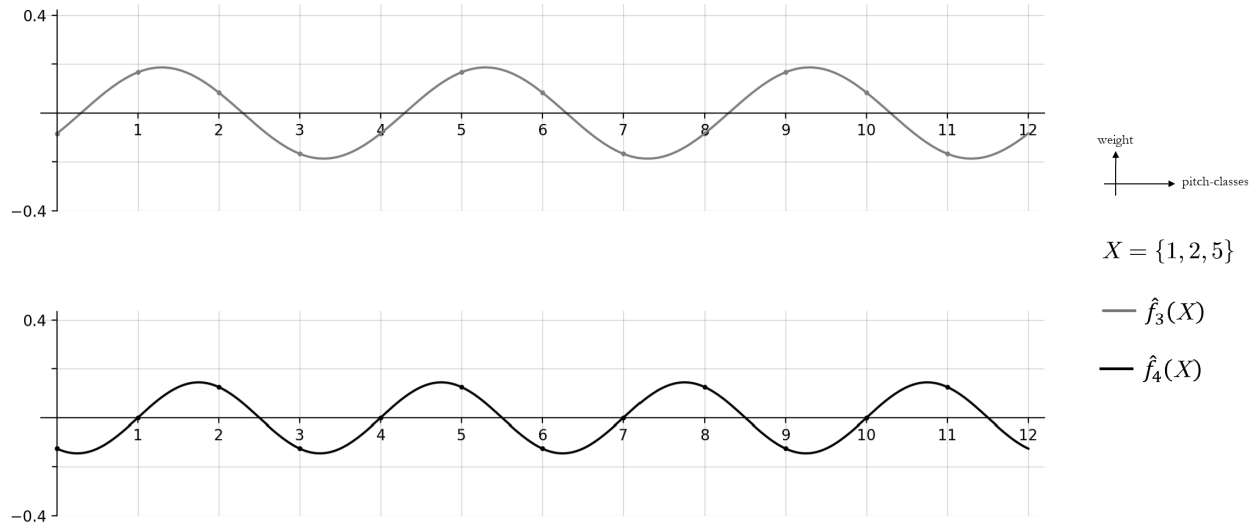
[28]Hereafter we will refer to set classes using Forte-No.[...] and pitch classes with {...}. For example, the pentachord formed by pitch-classes {2,4,5,6,8} belongs to set-class 5-8[02346].

[29]To obtain the DFT of a pcset X, $\hat{F}(X) = \left(\hat{f}_0, \hat{f}_1, \ldots, \hat{f}_k\right)$, one can calculate each DFT coefficient $\hat{f}_k$ given by $\hat{f}_k = \sum_{j=0}^{n-1} x_n e^{-i2\pi kj/n}$. In $\mathbb{Z}_{12}$, the coefficients range from 0 to 11 (Amiot [2016]; Yust [2015a]).

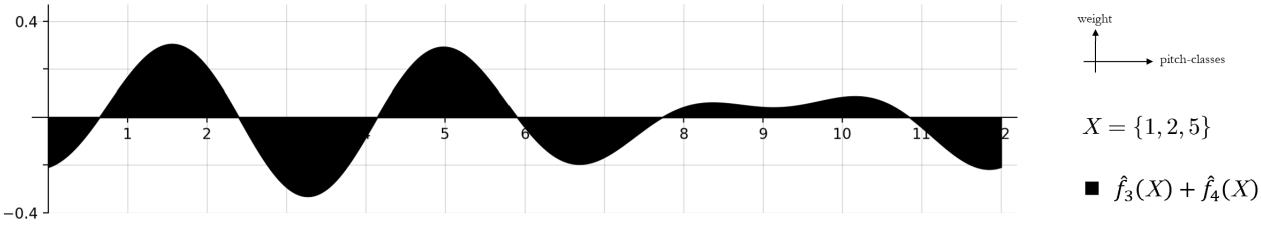
[30]One can determine DFT magnitudes and phases with $\left|\hat{f}_k\right| = \sqrt{a\left(\hat{f}_k\right)^2 b + \left(\hat{f}_k\right)^2}$ and $\varphi_k = \arctan\left(a\left(\hat{f}_k\right), b\left(\hat{f}_k\right)\right)$, respectively.

[31]To calculate the sinusoidal function of the k-th DFT coefficient please use the following equation $y(x) = \left|\hat{f}_k\right| \cos\left(2\pi \frac{k}{n} x + \varphi_k\right)$, where $\left|\hat{f}_k\right|$ is the sinusoid's amplitude, $\frac{k}{n}$ its frequency, and φ_k its horizontal shift (phase).

[32]Yust (2015b, 130).



EXAMPLE 4(a). *Periodic signals resulting from DFT coefficients* $\hat{f}_3$ *(in gray) and* $\hat{f}_4$ *(in black) for pcset* $X = \{1, 2, 5\}$.



EXAMPLE 4(b). *Graph depicting the sum of the DFT coefficients* $\hat{f}_3$ *and* $\hat{f}_4$ *for pcset* $X = \{1, 2, 5\}$.

One of the DFT's fundamental characteristics is that transpositions do not affect the DFT magnitudes but do affect phases.[33] Example 5 shows $\hat{F}(X)$ and $\hat{F}(Y)$, with Y transposed (T_6) from X. It shows that: (1) X and Y have equal magnitudes and therefore, DFT magnitudes are independent of transpositions and (2) X and Y denote a horizontal shift in their graphs, thus exposing that DFT phases are transposition dependent.

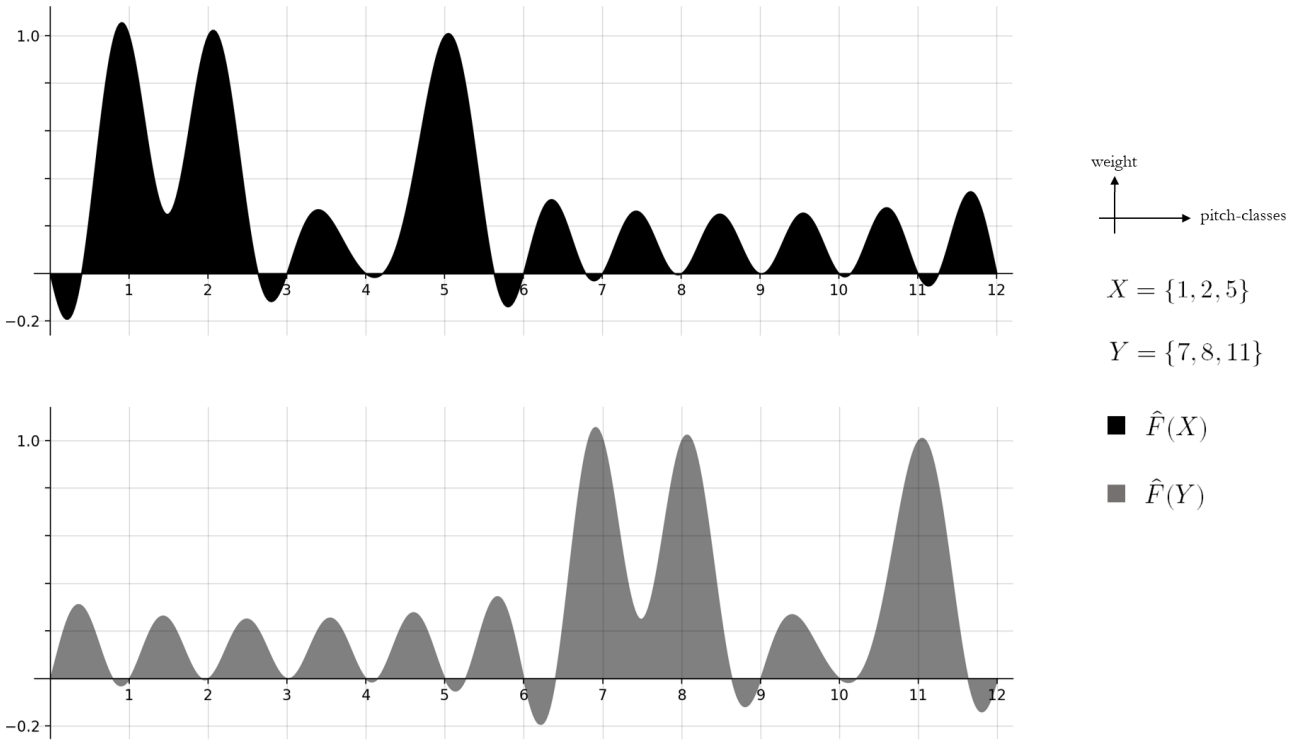
We already mentioned that $\hat{f}_0$ is the pcset's cardinality—e.g., $\hat{f}_0 = 3$ for trichord {1,2,5}. In the remaining DFT coefficients, the first six, $\hat{f}_1, \ldots, \hat{f}_6$, always have equal magnitude and opposite phase to $\hat{f}_{11}, \ldots, \hat{f}_7$. As such, $\hat{f}_1, \ldots, \hat{f}_6$ are the non-trivial DFT coefficients.[34]

Different interval classes and prototypical harmonic qualities relate to the non-trivial DFT coefficients.[35] Example 6 details the relation between $\hat{f}_1, \ldots, \hat{f}_6$ with chromaticity, dyadicity, triadicity, octatonicity,

[33]Amiot (2016, 11); Yust (2015b, 2016).

[34]Quinn (2007); Yust (2015b).

[35]The DFT coefficients and interval classes relate according to the following correspondence: $\hat{f}_1 \mapsto 1$, $\hat{f}_2 \mapsto 6$, $\hat{f}_3 \mapsto 4$, $\hat{f}_4 \mapsto 3$, $\hat{f}_5 \mapsto 5$, and $\hat{f}_6 \mapsto 2$ (Amiot [2017b]; Quinn [2007]; Yust [2015a]).



EXAMPLE 5. *Two graphical representations showing* $\hat{F}(X)$ *and* $\hat{F}(Y)$*, being* $X = \{1, 2, 5\}$ *and* $Y = \{7, 8, B\}$*. The horizontal shift suggests that* $\left|\hat{F}(X)\right| = \left|\hat{F}(Y)\right|$ *and* $\varphi(X) \neq \varphi(Y)$*.*

diatonicity and whole-tone quality, respectively.[36] The larger $|\hat{f}_k|$, the more features its pcset will exhibit from the quality associated with $\hat{f}_k$.

The sinusoidal representation of DFT coefficients helps to illustrate the connection between DFT magnitudes and harmonic qualities. Examples 7(a), 7(b), and 7(c) show six hypothetical sinusoids, representing DFT coefficients $\hat{f}_1, \ldots, \hat{f}_6$. These sinusoids are hypothetical since they are mere representations of the DFT coefficients' periodicity, and their magnitudes are indicative only. As such, each part of Example 7 presents sinusoids with periods per octave equal to those of DFT coefficients $\hat{f}_1, \ldots, \hat{f}_6$. One can understand $\hat{f}_1, \ldots, \hat{f}_6$ as measurement units for: (1) chromaticity, since $\left|\hat{f}_1\right|$ is higher for pcsets that concentrate on a specific segment of the pitch-class circle, as long as their cardinality is less or equal to six; (2) dyadicity, since the sinusoid

[36]Although there is a generic consensus in literature regarding the association of harmonic qualities with DFT coefficients, there are nuances in the terminology used. These nuances arise from the fact that DFT coefficients measure slightly different harmonic properties depending on their phase φ_k and that authors tend to focus their analyses on specific properties. For instance, in Yust's (2019) work, $\hat{f}_3$ is an indicator of triadicity, while in Chiu's (2021) article, the same coefficient indicates hexatonicity. Similarly, $\hat{f}_1$ can be a measure of chromaticity (Yust [2016], [2019]) or clustered quality (Chiu [2021]). To address these differences, we propose a nomenclature that is as generic and uniform as possible, as presented in Ex. 5. Specifically, we use the terms chromaticity, dyadicity, and triadicity for $\hat{f}_1$, $\hat{f}_2$, and $\hat{f}_3$, respectively, instead of clustered quality, quartal quality, and hexatonicity. This choice highlights the periodicity of the DFT coefficients, which ensures their association with different harmonic qualities.

Coefficient	Harmonic quality	Associated interval class	Specific features
$\hat{f}_1$	Chromaticity	IC1	Evaluates how structured the pcset is around a segment of the pc circle;
$\hat{f}_2$	Diadicity	IC6	Exhibits a higher value for pcsets with up to three chromatic tritones;
$\hat{f}_3$	Triadicity	IC4	Ranks higher for augmented triads or pcsets close to an hexatonic scale;
$\hat{f}_4$	Octatonicity	IC3	Receives higher values for 7th diminished chords and pcsets close to an octatonic scale;
$\hat{f}_5$	Diatonicity	IC5	Assesses the organization level in the pcset around a section of the circle of 5ths;
$\hat{f}_6$	Whole-tone quality	IC2	Evaluates how much the pcset is based in a whole-tone scale.

EXAMPLE 6. *Table associating the six nontrivial DFT coefficients, $\hat{f}_1, \ldots, \hat{f}_6$, with their respective harmonic quality, interval class, and interval properties.*

divides the octave into two periods—one period per tritone—and exhibits higher $\left|\hat{f}_2\right|$ for pcsets consisting of up to three tritones at a second minor distance;[37] (3) triadicity or hexatonicity, as the value of φ_3 allows $\hat{f}_3$ to measure how close a pcset is to an augmented triad or to the hexatonic scale; (4) octatonicity, weighing $\left|\hat{f}_4\right|$ how close the pcset is to a diminished seventh chord or to one of the two octatonic scales, depending on the value of φ_4; (5) diatonicity, since $\left|\hat{f}_5\right|$ captures the positioning of the pcset in the circle of fifths, and can also, in some contexts, represent its similarity to a pentatonic set; and (6) whole-tone quality, as its sinusoid splits the octave into six periods, and, depending on the value of φ_6, $\left|\hat{f}_6\right|$ measures how close the pcset is to one of the two whole-tone scales.

The DFT has allowed the development of geometric spaces in which pcsets become harmonic objects, i.e., points with a precise position in space.[38] These spaces provide advantages to analytical practice, such as objectively assessing the relative distance between pcsets or enabling the visualization of harmonic motion as trajectories in space.[39]

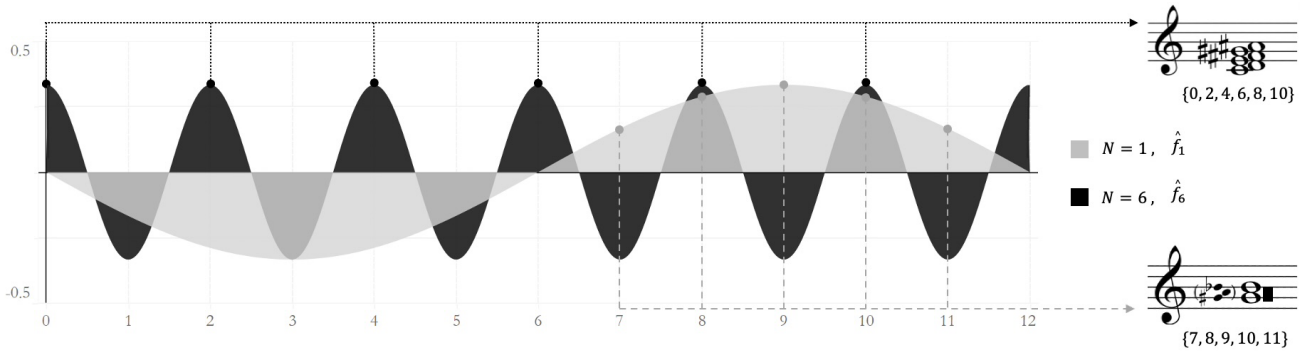
Quinn's work is one of the first to approach this matter. It culminates in quality space, a spatial framework that uses the link between DFT coefficients and harmonic qualities to illustrate how the opening of Ligeti's *Lux Aeterna* progressively moves from chromaticity ($\hat{f}_1$) towards diatonicity ($\hat{f}_5$).[40] This trajectory is evident in a Cartesian plane where the *x*-axis and *y*-axis refer to two DFT coefficients' magnitudes.

[37]In this regard, Messiaen's fifth mode of limited transposition presents this profile as it comprises pitch-classes {0,1,5,6,7,B}.

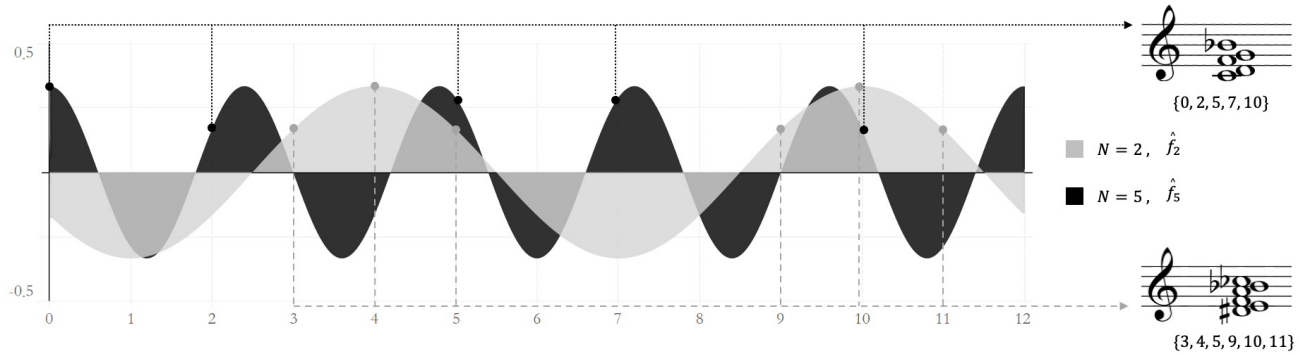
[38]In addition to the previously referenced examples by Amiot (2013), Quinn (2006, 2007), and Yust (2015b), other significant studies in which chords are treated as harmonic objects with precise positioning in a geometric space include the works of Hoffman (2008) and Tymoczko (2008, 2020).

[39]Yust (2015b, 121).

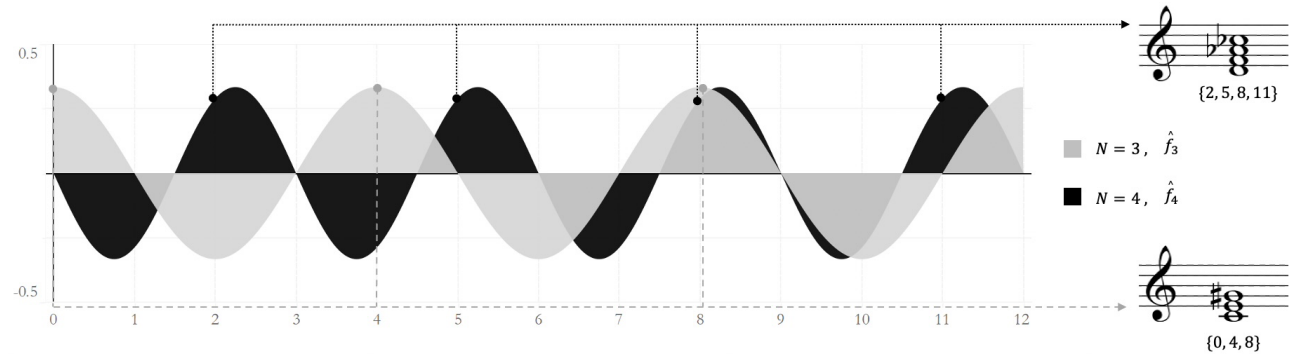
[40]Quinn (2007, 54).



EXAMPLE 7(a). *Representation of two hypothetical sinusoids where N represents the number of periods per octave, thus matching the periodicity of DFT coefficients. The positive segments of the sinusoids show distinct pcsets that help to illustrate how DFT coefficients capture the qualities listed in Example 6. Specifically, in this example, one has hypothetical sinusoids for* $\hat{f}_1$ *and* $\hat{f}_6$ *and their association with chromaticity and whole tone quality, respectively.*



EXAMPLE 7(b). *Depiction of two more hypothetical sinusoids, in this case, relating to* $\hat{f}_2$ *and* $\hat{f}_5$*. These coefficients capture dyadicity and diatonicity qualities.*



EXAMPLE 7(c). *Rendition of the final two hypothetical sinusoids relating with the six non-trivial DFT coefficients. In detail, this figure depicts sinusoids for* $\hat{f}_3$ *and* $\hat{f}_4$*, representing triadicity and octatonicity.*

DFT phases also allow one to visualize harmonic motion. Emmanuel Amiot and Jason Yust detail the spatial specificities resulting from the disposal of two DFT phases—φ_3 and φ_5, for example—in a Cartesian plane.[41]

[41]The space resulting from the works of Amiot (2013; 2016, 159–69) and Yust (2015b), constructed using the phases φ_3 and φ_5, exhibits a geometric structure similar to the Tonnetze, in which the different pitch classes cyclically form triangular structures.

Phase spaces are often two-dimensional reductions of a torus since φ_k has a cyclic character, and therefore, any space $\varphi_{i/j}$ is the product of two circles. The most suitable combination in functional harmony is phase space $\varphi_{3/5}$, using coefficients $\hat{f}_3$ and $\hat{f}_5$ for the analysis of Schubert's music, although other combinations for twentieth-century composers have been used.[42] Recently, Gilberto Bernardes, Nádia Carvalho, and Samuel Pereira proposed distorting Fourier space coefficients ($\varphi_{1\mapsto 6}$), visualized in two dimensions using the *Multidimensional Scaling* (MDS), to define a harmonic stability indicator.[43] These contributions are directly relevant to the objectives of our study as they seek to establish a metric for measuring harmonic stability (as opposed to ambiguity) and have been primarily evaluated in functional tonality (to the detriment of free atonality).

1.2 MAILMAN'S REPRESENTATIONAL HIERARCHY ASSOCIATION MODEL

To analyze the harmonic flow in Schoenberg's atonal works, Mailman proposes an analytical method he calls *Representational Hierarchy Association* (please refer back to the diagram in Example 2).[44] Mailman's methodology seeks to study how non-consecutive harmonic events relate in Schoenberg's atonal music, perceiving relationships of affinity or opposition between them. Consequently, these non-consecutive pcsets constitute a higher hierarchical level in the harmonic skeleton of a piece, as if gaining structural significance. Briefly, Mailman's methodology includes defining (1) criteria to segment a piece into smaller parts (phrase-segments), (2) criteria to select one pcset to represent each phrase-segment (a reference sonority, or RS), and (3) an abstract plane where the different phrase-segments associate through the RSs that represent them (an association arena).

In addition, Mailman states that the *Representational Hierarchy Association* is a flexible analytical model since it is possible to use various criteria to segment a work into phrase-segments and to select peak-point sonorities For example, Mailman uses silence (pauses) and slur marks as segmentation possibilities. For the selection of peak-point sonorities, he uses criteria such as the pcset with the highest note of a phrase-segment or the pcset coinciding with a melodic peak. Moreover, the author suggests the feasibility of set class genera (diatonic, hexatonic, and octatonic) and Fourier spaces as arenas in which peak-point sonorities can associate, a task we intend to tackle.

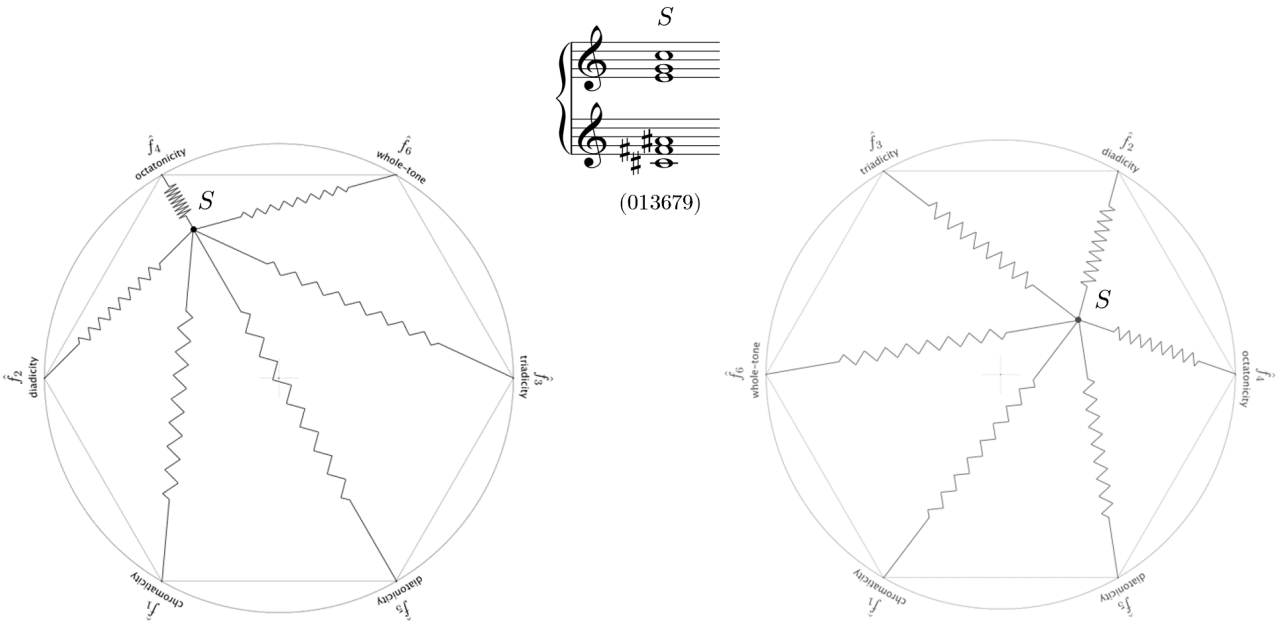
2 TOWARDS A TWO-DIMENSIONAL FOURIER QUALITY SPACE

The FQS projects $\mathbb{Z}_{12}$ pcsets as points inside a hexagon. It is split into six equal parts related to the six qualities of Example 6. Due to the properties of DFT coefficients, all pcsets sharing the same set class have equal

[42]For an analytical approach to Schubert's harmonic language with a $\varphi_{3/5}$ phase space, please refer to Yust (2015b). For analyses of twentieth-century music using different phase space combinations, see Pereira and Bernardes (2020) and Yust (2015a, 2016).

[43]Bernardes, Carvalho, and Pereira (2023).

[44]Mailman (2015).



EXAMPLE 8. *Illustration depicting the RadViz spring model applied to DFT coefficients and Stravinsky's "Petrushka" chord (represented with the letter S). The non-trivial DFT coefficients of* $\hat{F}(S)$ *are* $\left|\hat{f}_1\right| = 0$; $\left|\hat{f}_2\right| = 2$; $\left|\hat{f}_3\right| = 0$; $\left|\hat{f}_4\right| = 3.464$; $\left|\hat{f}_5\right| = 0$; *and* $\left|\hat{f}_6\right| = 2$. *The difference in S's position comes from the fact that there are two distinct arrangements of DFT coefficients along the circles. Reading in an anticlockwise direction and starting at the hexagon's rightmost vertex,* $\hat{f}_3, \hat{f}_6, \hat{f}_4, \hat{f}_2, \hat{f}_1, \hat{f}_5$ *is the arrangement in the FQS at the left and* $\hat{f}_4, \hat{f}_2, \hat{f}_3, \hat{f}_6, \hat{f}_1, \hat{f}_5$ *in the right.*

geometric coordinates in FQS, regardless of their pitch classes. The *RadViz* multidimensional data visualization technique develops the proposed analytic space.

2.1 THE FOURIER QUALIA SPACE

The FQS results from applying the *RadViz* technique to the DFT coefficients' magnitude. *RadViz* uses a unit circle where d dimensions interact through d equidistant points along the circumference. These points function as anchors and delineate a regular polygon with d sides.[45] The anchors in the FQS are the non-trivial DFT coefficients $\hat{f}_k$, with $k \in \{1, 2, 3, 4, 5, 6\}$, meaning that it considers six dimensions. Thus, the resulting space is a hexagon. Each DFT coefficient has two-dimensional coordinates $\hat{f}_k = (x_k, y_k)$ in the FQS. The *RadViz* technique then uses a physical-springs model to determine the 2-D coordinates of any set class (a six-dimensional DFT magnitudes object). Example 8 illustrates the FQS's hexagonal shape and the springs' attachment to DFT coefficients.

Consider now S, the set-class 6-30[013679], also visible in Example 8, corresponding to the *Petrushka* chord. This set class has six non-trivial DFT coefficient magnitudes, such that $S = \left(\left|\hat{f}_1\right|, \left|\hat{f}_2\right|, \ldots, \left|\hat{f}_k\right|\right)$. In FQS, the $\left|\hat{f}_k\right|$ value represents the stiffness of the k springs. Thus, the coordinates of S in the FQS are the 2-D point $P_S =$

[45] Dzemyda (2013, 9–12); Rubio-Sanchez et al. (2016).

(x, y), where the sum of the spring forces equals 0.[46] As such, one can conclude that: (1) the higher $\left|\hat{f}_k\right|$ is, the closer a set class will be to $\hat{f}_k$; (2) if $\left|\hat{f}_1\right| = \left|\hat{f}_2\right| = \cdots = \left|\hat{f}_k\right|$, then the set class will be positioned at the center of the FQS; and (3) if a set class only has one element such that $\left|\hat{f}_k\right| \neq 0$, then its position coincides with $\hat{f}_k$.

The order of DFT coefficients in the circle is a fundamental aspect of the FQS, since it ultimately determines the final positioning of a pcset in space. For instance, in Example 8, pcset S has a different position depending on the ordering of DFT coefficients. Note that S has $\left|\hat{f}_1\right| = 0$, $\left|\hat{f}_2\right| = 2$, $\left|\hat{f}_3\right| = 0$, $\left|\hat{f}_4\right| \approx 3.5$, $\left|\hat{f}_5\right| = 0$, $\left|\hat{f}_6\right| = 2$, explaining its final position close to the $\hat{f}_4$ anchor in both situations.

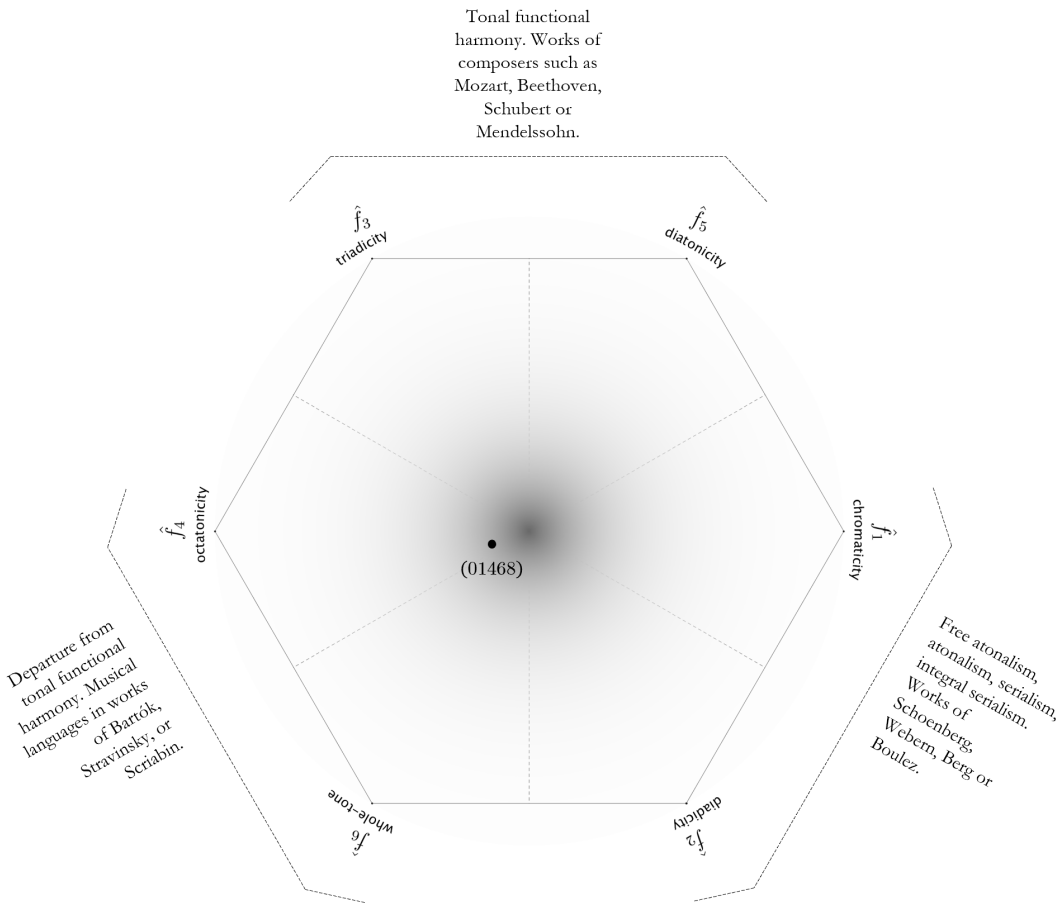
To define a specific order, we pursued a twofold strategy that aims at (1) favoring a greater dispersion of pcsets and (2) preserving the highest correlation between the positioning of pcsets in the FQS and their positioning in a six-dimensional space. We aim to provide a projection that distinguishes pcsets' qualities with maximal dispersion and preserves their distances. To this end, we performed two tests. First, we studied the positioning of all set classes for each circle's $6! = 720$ anchor permutations.[47] We plotted all set classes in space for each permutation and summed their Euclidean distances to the center. The permutation that maximized the sum was $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6, \hat{f}_4$.

Second, we assessed which anchor permutation best correlates with the six-dimensional DFT magnitude space to ensure that the FQS preserves distances with it. To this end, we computed the Pearson correlation coefficients between a matrix of pairwise Euclidean distances amongst all 220 set classes in six-dimensional space and corresponding distance matrices in two-dimensional FQS for each anchor permutation, thereby identifying the permutation that best preserves the 6-D spatial relationships. Orders $\hat{f}_3, \hat{f}_4, \hat{f}_6, \hat{f}_2, \hat{f}_5, \hat{f}_1$ and $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6, \hat{f}_4$ had the highest Pearson correlation coefficient, namely $r = .729$, which stands for a strong correlation. Note that $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6, \hat{f}_4$ is equivalent to the order resulting from the first test. Hence, we considered $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6, \hat{f}_4$ as the final order of DFT coefficients in FQS.[48] It is important to emphasize that

[46]To determine the point where the sum of all spring forces equals zero, please apply the following calculation $P_S = \left(\frac{\sum_{k=1}^{6} x_k \cdot \left|\hat{f}_k\right|}{\sum_{k=1}^{6} \left|\hat{f}_k\right|}, \frac{\sum_{k=1}^{6} y_k \cdot \left|\hat{f}_k\right|}{\sum_{k=1}^{6} \left|\hat{f}_k\right|}\right)$.

[47]We considered the 220 set classes in Straus's list (Straus [2016, 378–81]). Moreover, note that in P_6 permutations, there are equivalent possibilities. For instance, $\hat{f}_1, \hat{f}_2, \hat{f}_3, \hat{f}_4, \hat{f}_5, \hat{f}_6$ and its retrograded permutation $\hat{f}_6, \hat{f}_5, \hat{f}_4, \hat{f}_3, \hat{f}_2, \hat{f}_1$, are related arrangements since they produce equal distances between set classes in space.

[48]We recall that these tests aim to find the ordering of DFT coefficients in the FQS that ensures the greater dispersion among set classes and guarantees the highest reciprocity between the FQS and the six-dimensional space. But why it is specifically the combination $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6$, and $\hat{f}_4$ that ensures such conditions? One hypothesis lies in the interval properties of the data used for conducting the tests. When summing the interval-class vectors of the 220 set classes, one obtains ⟨659,668,661,672,659,339⟩. There is a noticeable balance among interval-classes 1, 2, 3, 4, and 5, except for interval-class 6, which appears less. Therefore, the tests suggest an ordering of DFT coefficients in space that groups odd coefficients in one hemisphere and even in the other since the even coefficients encompass tritones, and the odd ones do not. In other words, to ensure the highest possible correlation, it is natural that the tests result in grouping odd and even coefficients separately in space due to the lower prevalence of tritones in the set classes.

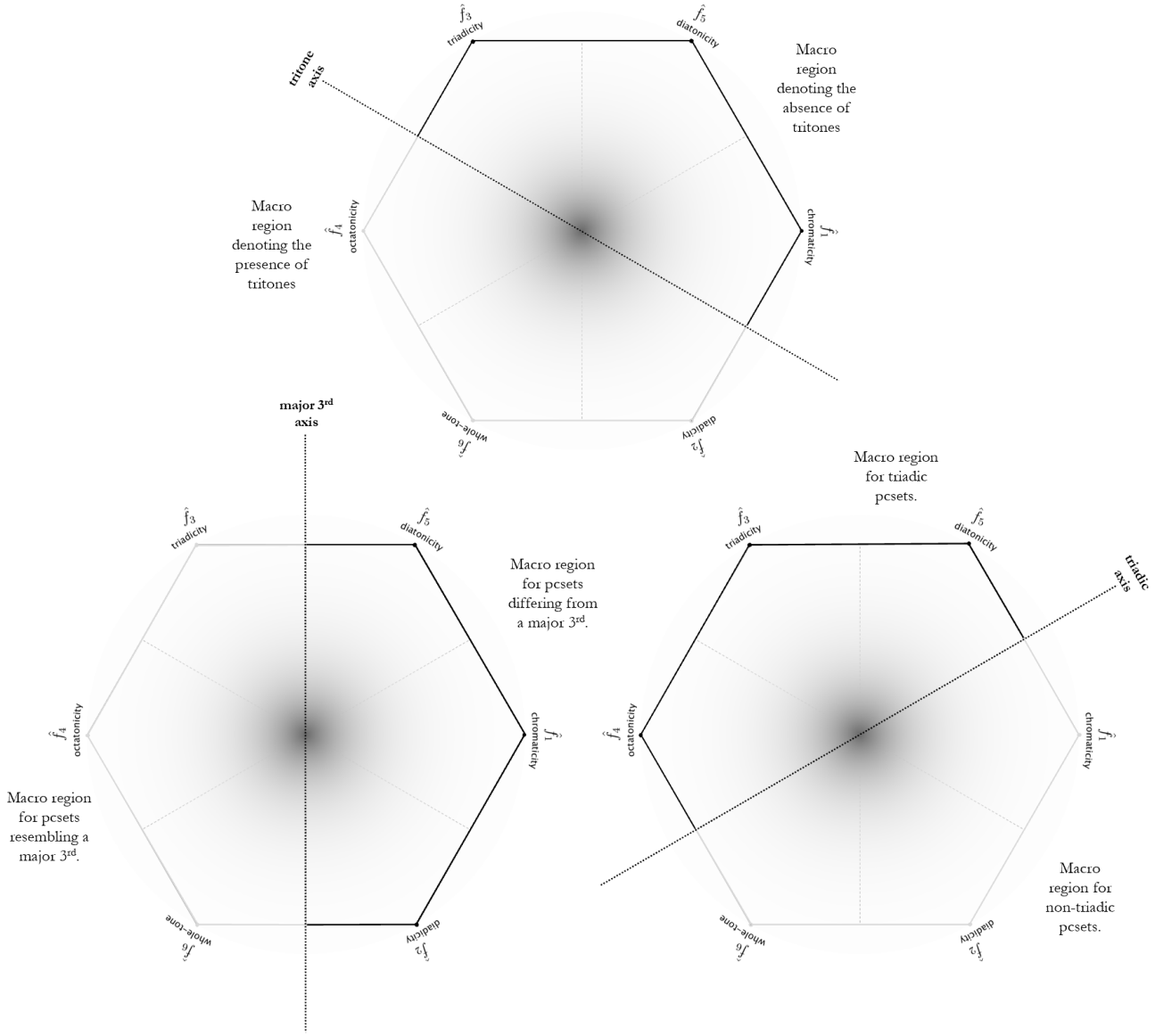


EXAMPLE 9(a). *Depiction of six regions in the FQS, each corresponding to a DFT coefficient, and three groups of regions linked to music with different pitch organizations. Finally, the example highlights the positioning in FQS of set class 5-30[01468]. Due to FQS's convex structure, its position is an interaction between all qualities. The center is the point of higher qualia ambiguity.*

a large part of the 27% distortion relates to the fact that the center of the space serves as an equilibrium point among DFT coefficients. In other words, points with equal magnitude value in each DFT coefficient—such as $(0, 0, 0, 0, 0, 0), (1, 1, 1, 1, 1, 1), \ldots$—will be distinct points in six-dimensional space, but they will have the same location in the FQS, namely, the center of the space. This characteristic stems from the FQS being a convex space, which imparts a certain narrowing around its center.

2.2 (FUZZY) REGIONS, CONVEXITY, AND "SINGULARITY"

It is vital to delve deeper into the idea with which we concluded the previous point, i.e., the possibility of outlining different DFT coefficient groups in the FQS that divide it into distinct regions, each with its own musical interpretation. The first and most evident way to do this is to delineate six regions whose area implies higher saliency in its respective DFT coefficient. To achieve this, we traced lines between the hexagon's center



EXAMPLE 9(b). *The three possible axes that divide the FQS into two equal parts are a tritone axis, a major third axis and a triadic axis. Each axis establishes two related macro-regions with intrinsic intervallic properties.*

and the midpoint of each of its six sides. Example 9(a) shows the resulting regions.[49] They provide a more intuitive interpretation of a set class position, easing the perception of its most preponderant qualities. Furthermore, one should note the possibility of identifying three distinct groups of regions, each with two harmonic qualities. Such regions suggest music with distinct pitch organizations. The first group joins $\hat{f}_5$ and $\hat{f}_3$, associated with functional tonal harmony. The second unites octatonicity and whole-tone quality, which mark the early twentieth-century repertoires of composers such as Stravinsky and Bartók. The third embraces dyadicity and chromaticity, fundamental to twentieth-century music, such as integral serialism. Example 9(a) also illustrates these three larger regions.

[49]The gradually transparent lines are related to increasing qualia ambiguity in the central zone of FQS, a subject addressed in detail in the following pages.

However, we can still design a new layer of regions with analytical meaning. For this, dividing the FQS into groups of three DFT coefficients is convenient through the vertical and oblique axes of the previous regions. Thus, it is possible to interpret the positioning of a pcset or set class in light of the following binaries: (a) $\hat{f}_3$, $\hat{f}_5$, $\hat{f}_1$ vs. $\hat{f}_2$, $\hat{f}_6$, $\hat{f}_4$, (b) $\hat{f}_5$, $\hat{f}_1$, $\hat{f}_2$ vs. $\hat{f}_6$, $\hat{f}_4$, $\hat{f}_3$, and (c) $\hat{f}_1$, $\hat{f}_2$, $\hat{f}_6$ vs. $\hat{f}_4$, $\hat{f}_3$, $\hat{f}_5$. Example 9(b) presents these divisions of the FQS schematically. The axis that divides the upper combination separates the even and the odd DFT coefficients and thus indicates the presence/absence of tritones. The combination on the left shows how much a set class resembles a major third, therefore implying high $\left|\hat{f}_3\right|$ and $\left|\hat{f}_6\right|$ along with low $\left|\hat{f}_1\right|$, $\left|\hat{f}_2\right|$, and $\left|\hat{f}_5\right|$. Finally, the axis of combination on the right separates set classes of greater or lesser triadic character, i.e., triadic pcsets will tend to be positioned in the region formed by $\hat{f}_4$, $\hat{f}_3$, and $\hat{f}_5$, and non-triadic pcsets in the region formed by the DFT coefficients $\hat{f}_1$, $\hat{f}_2$, and $\hat{f}_6$. As such, although the FQS is a reduction of a six-dimensional space, the possibility of outlining the multiplicity of regions that we now describe and interpreting the positioning of set classes according to these regions gives the FQS its analytical value and differentiator.

The FQS proposes convex combinations between $d = 6$ anchors because it normalizes DFT coefficient magnitude values so that their sum equals unity. Thus, one can deduce that any pcset not positioned at one of the anchors will always be a convex combination between two or more DFT coefficients.

This property shows that the regions outlined in Examples 9(a) and 9(b) are fuzzy spaces, i.e., each region allows a multitude of convex combinations between the six qualities. Example 9(a) also depicts pentachord 5-30[01468] closely positioned to the center and the boundary between $\hat{f}_4$ and $\hat{f}_6$ regions. One might think that this set class is merely octatonic since it lies within the $\hat{f}_4$ region, but that idea would be misleading. The pentachord's final position is a convex combination of its DFT magnitudes: $\left|\hat{f}_1\right| = 0.52$, $\left|\hat{f}_2\right| = 1.73$, $\left|\hat{f}_3\right| = 2.23$, $\left|\hat{f}_4\right| = 1$, $\left|\hat{f}_5\right| = 1.93$, $\left|\hat{f}_6\right| = 3$. What happens is that the higher saliency in $\hat{f}_6$ approximates 5-30[01468] to the whole-tone area. However, the secondary saliency in $\hat{f}_3$ slightly moves its positioning towards triadicity, thus resulting in a final position inside the $\hat{f}_4$ region.

The FQS's center has the most significant qualia ambiguity since it implies a balance of forces between two or more DFT coefficients.[50] Its vicinities constitute a singular and heterogeneous zone where two or more qualities reside with similar intensity. Consequently, the closer a pcset is to the center, the more complex its qualia interpretation will be.[51] In Examples 9(a) and 9(b), the gradient progressively becomes darker as one

[50]Examples of set classes precisely positioned at the center of the FQS are individual pitch classes, set-class 11-1[0123456789A], and set-class 12-1[0123456789AB], which result in $\left|\hat{f}_k\right| = 1$ and $\left|\hat{f}_k\right| = 0$, respectively.

[51]The idea of a circular space with symbolic value at its center evokes the concept of "perfect balance" (Milne et al. [2015]). In a circle with the twelve pitch classes, any set of points whose average position is at the center of the circle is "perfectly balanced" (for example, the notes of an octatonic scale). The authors proposed even a mathematical metric to "balance" using the DFT coefficient $\hat{f}_0$. In the FQS, the balance between DFT coefficients $\hat{f}_1$ to $\hat{f}_6$ leads to an unclear interpretation of harmonic qualities and positionings close to the center of the space.

#	Pcset	SC	$\lvert\hat{f}_1\rvert$	$\lvert\hat{f}_2\rvert$	$\lvert\hat{f}_3\rvert$	$\lvert\hat{f}_4\rvert$	$\lvert\hat{f}_5\rvert$	$\lvert\hat{f}_6\rvert$
1	Magic-hexachord	(014589)	0	0	4.24	0	0	0
2	Diminished 7th	(0369)	0	0	0	4	0	0
3	Messiaen's 5th mode	(012678)	0	4	0	0	0	2
4	Chromatic tetrachord	(0123)	3.35	1.73	0	1	0.9	0
5	Chromatic pentachord	(01234)	3.73	1	1	1	0.27	1
6	Diatonic scale	(013568T)	0.27	1	1	1	3.73	1

EXAMPLE 10(a). *Name, set class, and DFT magnitude size values for six different pcsets.*

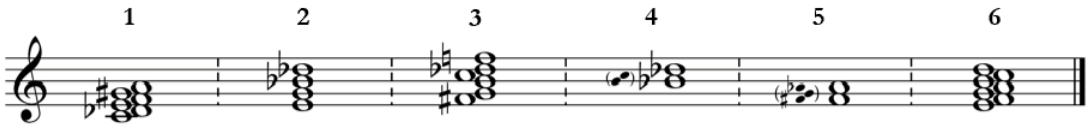
approaches the center and tries to express the idea of an increasingly ambiguous interpretation. With a similar rationale, the dashed lines progressively become transparent because the idea of multiple regions dilutes as one approaches the center.

However, the peculiarity of an ambiguous zone in space can be helpful. It enables comparing any two groups of set classes, assessing which one is more ambiguous, i.e., which is more proximate to the center. The ability to assess qualia ambiguity through the center of the space comes directly from the FQS and the mathematical properties of the *RadViz* technique and is not replicable in six dimensions. For example, a single note whose magnitudes are $\lvert\hat{f}_1\rvert = 1, \lvert\hat{f}_2\rvert = 1, \lvert\hat{f}_3\rvert = 1, \lvert\hat{f}_4\rvert = 1, \lvert\hat{f}_5\rvert = 1, \lvert\hat{f}_6\rvert = 1$, in the FQS, stands in the center of the space due to the balance between coefficients. Consequently, one can interpret this pcset as ambiguous. However, in six dimensions, the distance to the center, i.e., the point with coordinates $(0,\ 0,\ 0,\ 0,\ 0,\ 0)$, would not be an indicator of ambiguity.[52] In other words, the fact that the FQS funnels around its center is a unique property that allows for defining a metric for qualia ambiguity, a task that we address in Section 3.

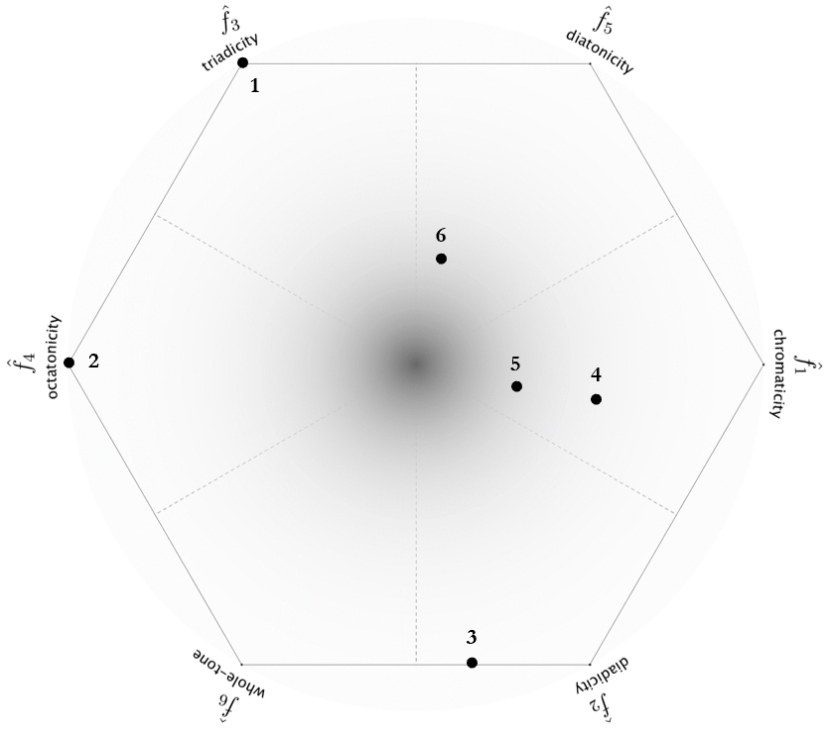
2.3 INTERPRETING SET CLASSES' POSITION IN FOURIER QUALIA SPACE

Example 10(a) lists six pcsets with high saliency in distinct DFT coefficients, and Example 10(b) shows one of their transpositions in the score. Their position in the FQS is visible in Example 10(c). One of these pcsets is the set class 6-20[014589], often called the *magic hexachord*. It is salient in $\hat{f}_3$ because it consists of two augmented triads a minor second apart. Quintessentially, this set class coincides with the positioning of the $\hat{f}_3$ anchor and is highly representative of triadicity. Similarly, the diminished seventh chord 4-28[0369] stands at the coordinates of the $\hat{f}_4$ anchor since it splits the octave into four equal parts.

[52]In this context, consider the point $p = (1, 1, 1, 1, 1, 1)$ and the center $q = (0, 0, 0, 0, 0, 0)$. The Euclidean distance between these two points is given by $d(p, q) = \sqrt{(1-0)^2 + (1-0)^2 + (1-0)^2 + (1-0)^2 + (1-0)^2 + (1-0)^2} = \sqrt{6}$. Similarly, the distance between the point $r = (2, 2, 2, 2, 2, 2)$ and the center is $d(r, q) = 2\sqrt{6}$. This result is inconsistent with a metric designed to geometrically quantify qualia ambiguity in relation to the center, as these Euclidean distances differ despite the balanced coordinates of p and r. Conversely, within the FQS, these points would be located at the center and exhibit maximum qualia ambiguity.



EXAMPLE 10(b). *One of the possible transpositions for the six pcsets in Example 10(a).*

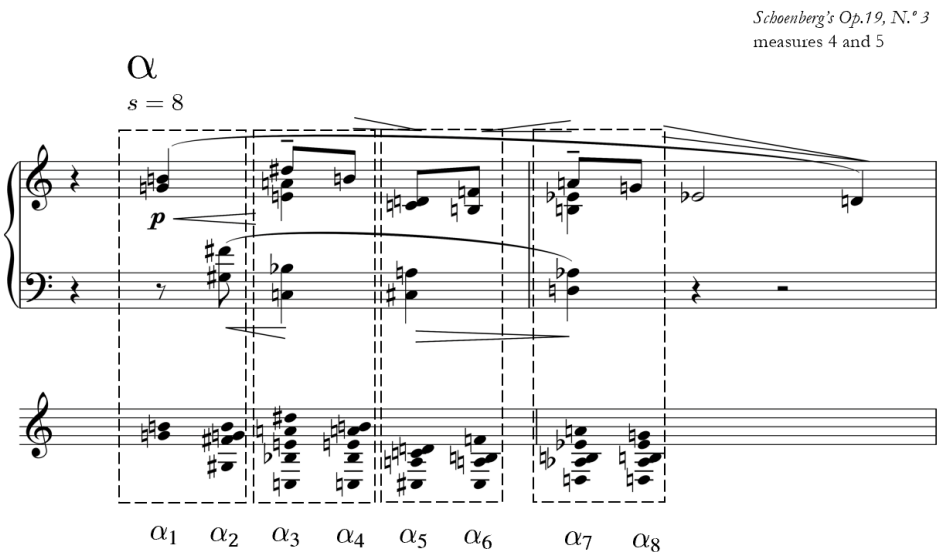


EXAMPLE 9(c). *The position in the FQS of the six pcsets in Example 10(a).*

Interpreting the remaining pcsets in Example 10(a) is more challenging due to their saliency in multiple qualities. The set-class 6-7[012678], known as Messiaen's fifth mode of limited transposition, has higher saliency in $\hat{f}_2$, followed by $\hat{f}_6$.[53] In Example 10(c), the pcset is closer to the $\hat{f}_2$ anchor with a slight displacement toward $\hat{f}_6$ due to the smaller spring force applied by $\hat{f}_6$. Therefore, by its mere positioning, one can deduce that this pcset presents a greater preponderance in dyadicity, tending, nevertheless, in the direction of whole-tone quality. The fact that pitch classes {1,5,7,B} are a subset of the whole-tone scale {1,3,5,7,9,B} explains this tendency.

On the other hand, one would think that the more semitones a pcset has, the higher its saliency in $\hat{f}_1$. However, as the pcset cardinality increases, its chromaticity level decreases. For example, 4-1[0123] and 5-1[01234] are both close to the $\hat{f}_1$ anchor, but the tetrachord is closer. In both pcsets, the magnitude in $\hat{f}_1$ is the most salient, but the residual magnitude in the remaining coefficients shifts their positioning away from the

[53]This set class has no magnitude in $\hat{f}_1$ because higher cardinalities can produce low $\left|\hat{f}_1\right|$, regardless of the number of semitones (Yust [2019, 219]). For example, the scale {0,1,4,5,7,8,B} results in $\left|\hat{f}_1\right| = 0$.



EXAMPLE 11(a). *The third phrase of Schoenberg's Op. 19, No. 3, mm. 4–5, represented as* α. *In this case* $s = 8$, *meaning that a total of eight set classes,* $\alpha_{1\mapsto 8}$, *are identifiable: 2-4[04], 4-4[0125], 5-19[01367], 5-5[01237], 4-4[0125], 4-24[0248], 5-19[01367], and 5-22[01478], respectively. These set classes result from all harmonic events in the phrase and the boxes help to illustrate how set classes* $\alpha_{1\mapsto 8}$ *are determined.*

$\hat{f}_1$ anchor. Finally, let us look at diatonicity. The position of heptachord 7-35[013568A] is towards $\hat{f}_5$, but its equal saliency in $\hat{f}_2$, $\hat{f}_3$, $\hat{f}_4$, and $\hat{f}_6$ exerts force so that it is not utterly coincident with the $\hat{f}_5$ anchor.

3 REFERENCE SONORITIES AND DISAMBIGUATING AMBIGUITY

This section presents an analytical method based on Mailman's *Representational Hierarchy Association.*[54] Our approach uses the FQS and its properties to define the criteria for selecting the RSs. Mailman notes that one could use other criteria, leading to different RSs and distinct analytical remarks. However, our goal is to explore the unique characteristics of the FQS, such as its different regions or the possibility of assessing qualia ambiguity. Thus, we formulate mathematical and computational criteria to reduce a piece to its most representative elements.

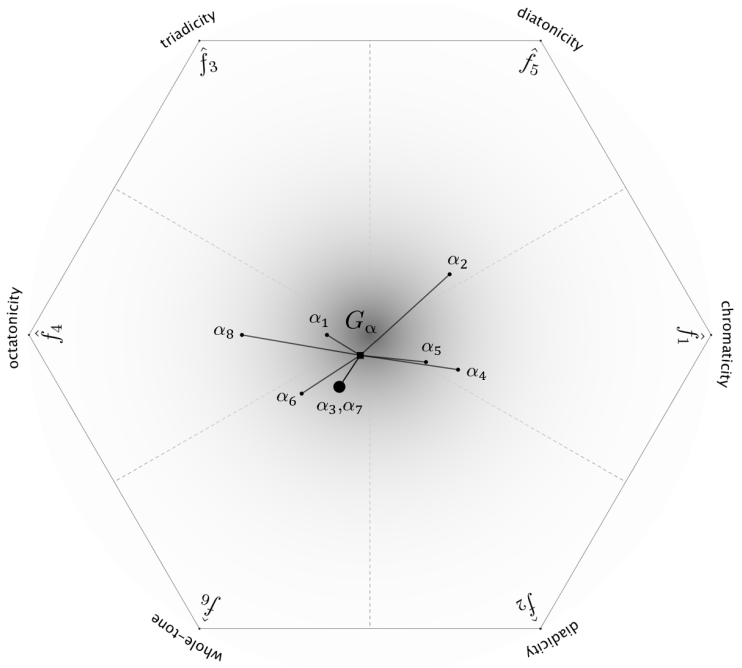
3.1 SEGMENTATION UNIT, REFERENCE SONORITIES, AND QUALIA MOTION

We chose the musical phrase as the segmentation unit for our *Representational Hierarchy Association* model. A musical phrase is a well-defined structural meaning unit, identifiable in Schoenberg's music due to its use of conventional formal structures.[55]

Consider a phrase α understood as a time series of *s* set classes. Each of the *s* set classes represents a harmonic event in the phrase, thus not considering individual pitch classes. Example 11(a) illustrates this process in mm. 4 and 5 of Schoenberg's Op. 19, No. 3. After determining the set classes' coordinates in the FQS, the point in space that best represents the qualia of α is its centroid (G_α), i.e., the linear interpolation of all its

[54]Mailman (2015).

[55]Boss (2019, 5).



EXAMPLE 11(b). *The position in the FQS of the eight pcsets* ($\alpha_{1\mapsto 8}$) *in the third phrase of Schoenberg's Op.19, No.3.*

constituents.[56] One can understand the centroid *G* as the ideal RS as it considers all pcsets equally. However, due to the continuous nature of the space, the resulting centroid may not coincide with a set class in α. For this reason, we considered for RS the set class that shares the most characteristics with G_α, i.e., the set class with the smallest Euclidean distance to the ideal sonority.[57]

Example 11(b) illustrates this method to determine an RS, showing the centroid G_α as a square and the RS_α as a large circle.[58] In this example, the RS_α is the set-class 5-19[01367], a fact already expected since it is the only set class heard twice in the phrase. Like a reductive analysis, all RSs in a work establish a high-level hierarchical progression that reduces the intricate harmonic web of the piece. One can also analyze such progression regarding voice leading, pitch range, interval structure, and the concrete trajectory it follows in the FQS.

3.2 MEASURING QUALIA AMBIGUITY

The FQS allows the assessment of ambiguity in harmonic qualia (η) for any musical phrase, measuring how linked its set classes are (or are not) to harmonic qualities. Thus, phrases with set classes interpretable over

[56]If (x_m, y_m) are the FQS's coordinates of the *s* pcsets in phrase α, its centroid coordinates are $G_\alpha = \left(\frac{1}{s}\sum_{m=1}^{s} x_m, \frac{1}{s}\sum_{m=1}^{s} y_m\right)$.

[57]It is possible to replicate the present method for choosing an RS in a six-dimensional space and only *a posteriori* project it in the FQS. However, given the FQS's unique characteristics—such as its different regions and center—we performed all calculations based on the FQS's (2-D) coordinates. Thus, we ensure that the FQS is the essence of the analytical approach we now advance and use throughout our remarks concerning Schoenberg's Op. 19, No. 1.

[58]In the following link, it is possible to find the Python code to plot in the FQS any group of pcsets and compute its centroid and RS: https://colab.research.google.com/drive/1MPJfpAr2Qp_CAnBJ-5_mzHQxvEOIRVzl?usp=sharing.

different qualities will have high ambiguity, whereas phrases formed by set classes strongly linked to any single quality will present low ambiguity.[59] As such, we measure qualia ambiguity by calculating the Euclidean distances of set classes to the FQS's center. Therefore, the η value will increase the less bound the set classes are to any harmonic qualities and the more concentrated around the center. Conversely, a low η will imply higher saliency in qualia and more dispersion relative to the center.

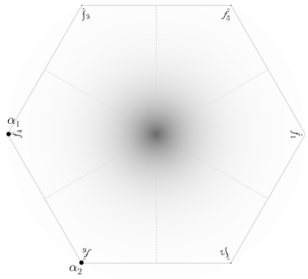
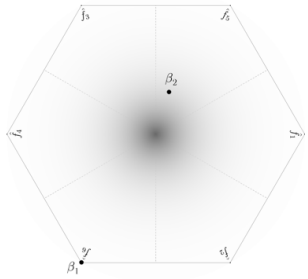
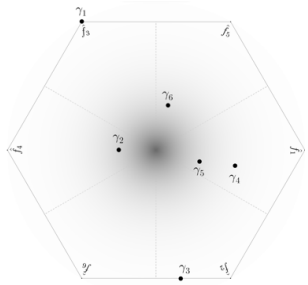
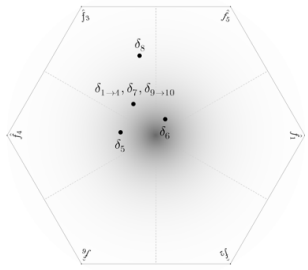
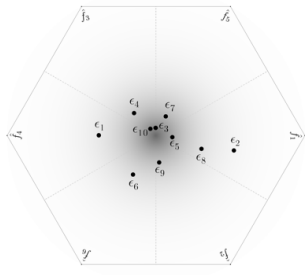
Consider now α as a generic group of *s* pcsets. Its qualia ambiguity, η_α, can be defined as one minus the mean of the Euclidean distances between the center of the space *c* and the *s* pcsets in α.[60] Thus, the ambiguity of α is given by $\eta_\alpha = 1 - \bar{l}_\alpha$, $\bar{l}_\alpha$ being the arithmetic mean of the *s* Euclidean distances to the center in group α. Please consider that we construct the FQS on a unit circle and, therefore, $0 \leq \eta_\alpha \leq 1$.

To illustrate η_α and demonstrate which ambiguity values are high, moderate, and low, we chose to study five groups of set classes. Example 12 summarizes our results and presents an increasing η for the five groups. The first three, α, β, and γ, are extreme examples with significantly reduced ambiguity values. Group α, for instance, consists of two pcsets with maximum salience in the harmonic qualities of octatonicity and whole-tone quality, and their position coincides with the $\hat{f}_4$ and $\hat{f}_6$ anchors. Thus, since the average of their Euclidean distances is maximum, then $\eta_\alpha = 0$. In the case of group β, the fact that 5-35[02479] is not a set class of maximum salience leads to an increase in ambiguity, resulting in $\eta_\beta = .347$. Group γ comprises the pcsets in Example 10(a) and receives a similar rationale. On the other hand, the δ and ϵ groups exhibit moderate to high ambiguity values, as expected for specific musical phrases. The first phrase of Bach's chorale *Befiehl du deine* Wege (BWV 244, No. 53) is an example that confirms that diatonic passages show high salience in $\hat{f}_3$ and $\hat{f}_5$ simultaneously.[61] Thus, δ justifies moderate η values. On the other hand, the ϵ group is an example of ten randomly generated pcsets and their respective set classes presenting the highest ambiguity value in Example 12.

[59]This idea resembles the metric of entropy (H) Amiot (2021) proposed. More specifically, entropy measures the interval organization of a pcset, i.e., a quantity that will be low for pcsets with strong organizational principles and high for pcsets with more random distributions (Amiot [2021, 5]). In other words, it assesses the internal organization of the distribution. The metric we propose for qualia ambiguity is distinct from entropy as it assesses the balance of forces between two or more DFT coefficients and measures how interpretable a group of set classes is considering any harmonic quality. In this regard, see α in Example 11, where $\eta_\alpha = 0$ since the two set classes are unequivocally associated with one of the harmonic qualities, but $H(0369) = .69$ and $H(02468A) = 0$, and therefore, the mean entropy is $\bar{H}(\alpha) = .345$. For better comparison between entropy and ambiguity, the average entropies for the other groups in Example 11 are: $\bar{H}(\beta) = .74$, $\bar{H}(\gamma) = 1.25$, $\bar{H}(\delta) = 2.08$, and $\bar{H}(\epsilon) = 1.96$.

[60]It is important to note that η_α subtracts the mean to one by the fact that the mean would only be a measure of objectivity and not of ambiguity. For example, it would increase the more bound set classes were to a given quality. As such, we subtract the mean to one to maintain the relation between ambiguity and its measurement unit.

[61]Matt Chiu (2021) argues that diatonicity arises particularly in pcsets exhibiting high salience in both DFT coefficients $\hat{f}_5$ and $\hat{f}_3$, rather than being merely a consequence of high salience in $\hat{f}_5$.

	Description	*Set classes*	η	*FQS positioning*
α	Consecutive FMS anchors	#1: (0369) #2: (02468T)	$\eta_{\alpha} = 0$	
β	Opposite pcsets in FMS	#1: (02468T) #2: (02479)	$\eta_{\beta} = .347$	
γ	Pcsets in Figure 6	#1: (014589) #2: (0369) #3: (012678) #4: (0123) #5: (01234) #6: (012568T)	$\eta_{\gamma} = .453$	
δ	Diatonic phrase	#1: (037) #2: (037) #3: (037) #4: (037) #5: (0258) #6: (0247) #7: (037) #8: (0358) #9: (037) #10: (037)	$\eta_{\delta} = .713$	
ϵ	Random pcsets	#1: (01478) #2: (01234567) #3: (0123469) #4: (014679) #5: (0123568) #6: (023468) #7: (01234578T) #8: (01234) #9: (016) #10: (0145679)	$\eta_{\epsilon} = .772$	

EXAMPLE 12. *Qualia ambiguity values for six distinct groups ($\alpha \rightarrow \epsilon$) of set classes and their respective position in the FQS.*

4 SYMMETRY, LINEARITY, AND AMBIGUITY IN SCHOENBERG'S OP. 19, NO. 1

The idea of linear motion governing pitch, voice leading, or meter is one of the fundamental aspects highlighted in Kramer's analytical interlude examining Schoenberg's Op. 19, No. 1. The author calls it

Phrase	Bars	s	SCs	Aggregated SCs
α	0 and 1	8	$\alpha_1 \mapsto \alpha_8$	$1-8$
β	2 to 4 (1st beat)	19	$\beta_1 \mapsto \beta_{19}$	$9-27$
γ	4 (2nd beat) to 6	20	$\gamma_1 \mapsto \gamma_{20}$	$28-47$
δ	7 to 12	25	$\delta_1 \mapsto \delta_{25}$	$48-72$
ϵ	13 and 14	8	$\epsilon_1 \mapsto \epsilon_8$	$73-80$
ζ	15 to 17	11	$\zeta_1 \mapsto \zeta_{11}$	$81-91$

EXAMPLE 13. *Phrase segmentation for the Op.19, No.1. We identified six phrases represented by α, β, γ, δ, ϵ and ζ. The example shows the phrase's bar numbers, the total number of set classes (s) per phrase, as well as the aggregated number of set classes identified in the piece. Set classes are abbreviated to SCs.*

"nondirected linearity" since it has no unequivocal goal.[62] Such linearity creates movement that propels the music forward as it unfolds, but the listener does not expect a clear resolution to the movement and tension created. Supported by the FQS and the method explained in Section 3, we tried to understand if this or other aspects arise in the qualia development of Schoenberg's Op. 19, No. 1.

4.1 THE REFERENCE SONORITIES FOR OP. 19, NO. 1

Example 13 and Example 14 show the segmentation of the piece into six phrases, α, β, γ, δ, ϵ and ζ, their respective RSs, and the ninety-one pcsets found in the example.[63] The silence at closure delimits phrases α, β, and δ. Phrase δ ends in m. 12 with a deceleration of events starting in m. 10, and the suspension reinforces δ_{25} as the phrase's final pcset. In m. 14, the deceleration in the work's tempo and the sudden register change mark a cadence that further enhances both the end of ϵ and the beginning of ζ.

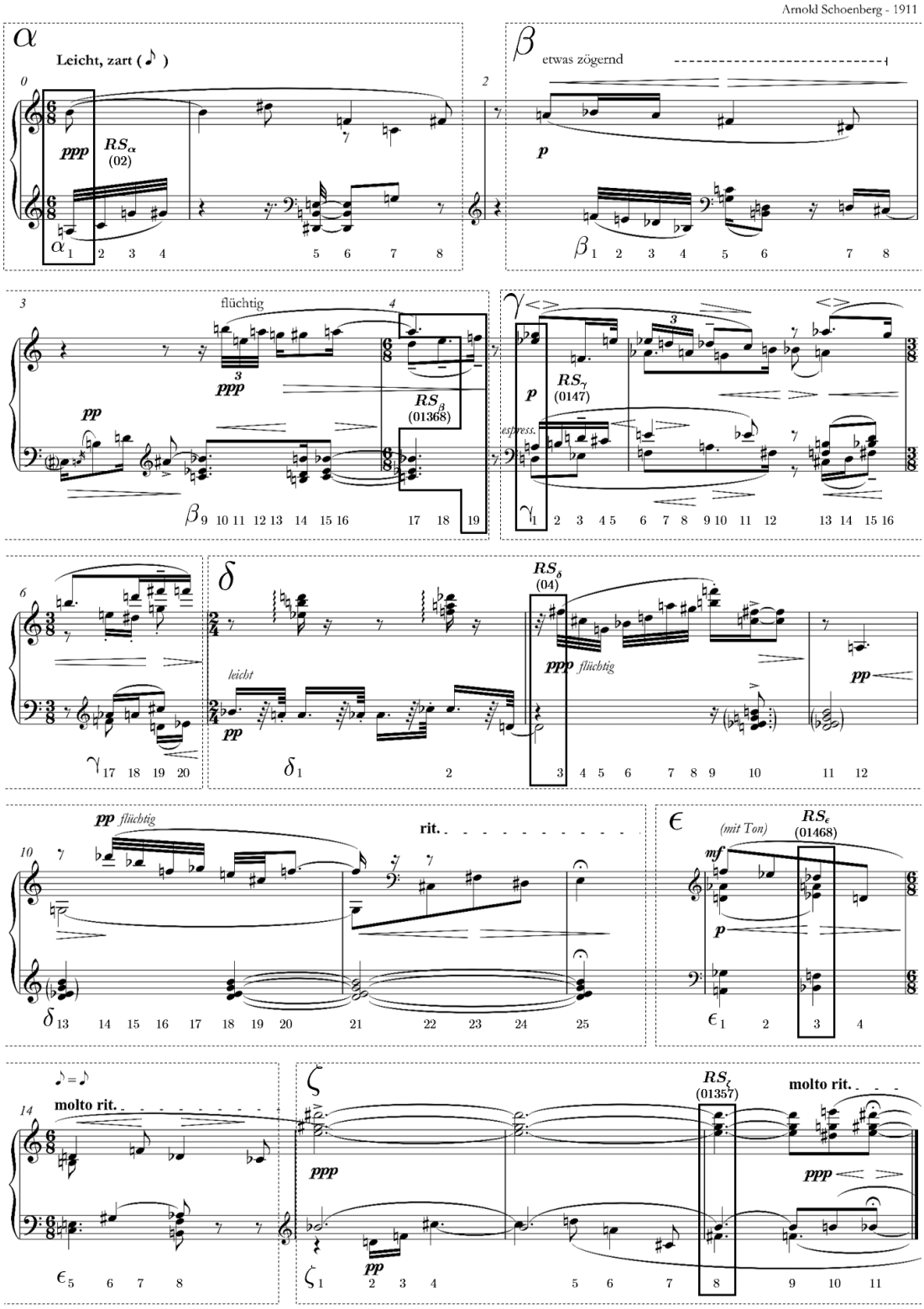
Finding a cadence between phrases γ and δ is not as evident as if silence separated them. However, the sudden textural change encouraged us to interpret it as a rupture with the preceding music.[64] The "contrapuntal" texture of the first six measures suddenly disappears, and a contrasting texture follows in m. 7, in which the left hand plays a melody of a short and articulate character. In contrast, the right hand plays two arpeggiated pcsets.

Examples 15(a), 15(b), and 15(c) show the six phrases, their centroids, and respective RSs in the FQS. The RS_α is the first pcset of the work, i.e., the dyad {9,B} ↦ 2-2[02]. This set class corresponds to interval-class 2, found in three moments: α_1, α_6 and α_7. In addition, interval-class 2 appears in three distinct forms: (1) as an interval

[62]Kramer (1988, 170–83).

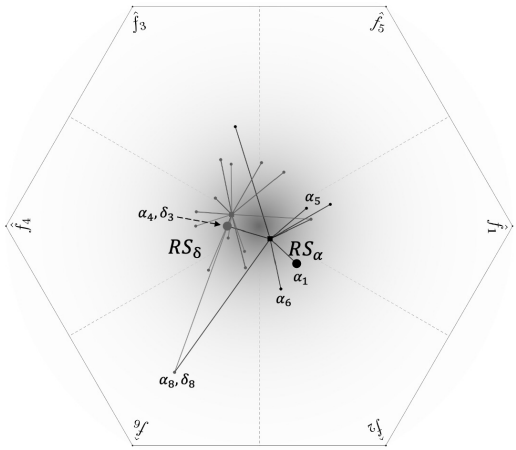
[63]We did not consider the sixth thirty-second note in m. 8 since it has only one pitch class. There is also the case of the tremolo beginning in m. 8, which articulates {2,3,7} and {B}. We opted to consider both as the single entity {2,3,7,B} since the tremolo is a textural artifice, and one perceives both sonorities simultaneously.

[64]Ashforth (1978, 220); Kramer (1988, 170–83).

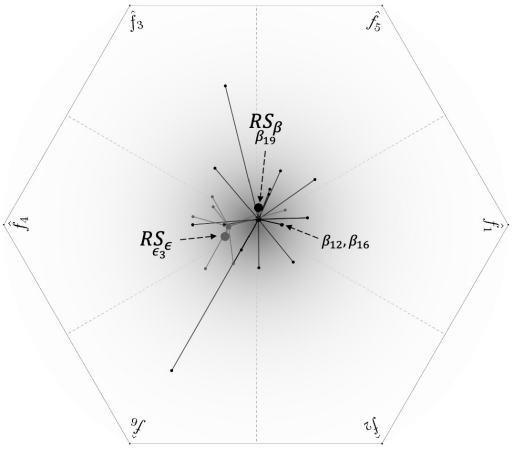


EXAMPLE 14. *Annotated transcription of Schoenberg's Op. 19, No. 1. We considered a total of 91 pcsets. Phrases* $\alpha \rightarrow \zeta$ *appear with dashed lines and the RSs are the pcsets* α_1, β_{19}, γ_1, δ_3, ϵ_3, *and* ζ_8 *marked with solid lines.*

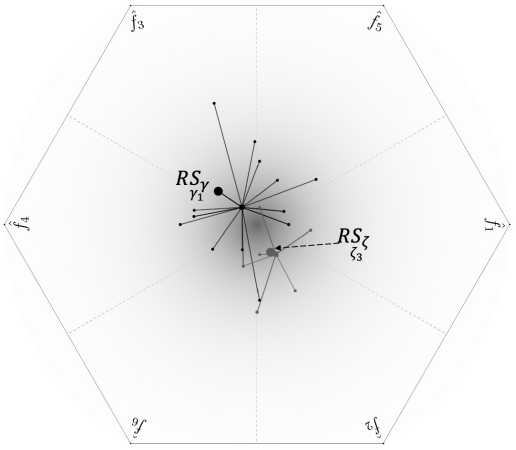
in α_1, (2) in melodic form with the minor seventh leap between D♯ and F in the right hand, and (3) blended in α_6 and α_7. These ideas suggest an appearance of interval-class 2 with increasing complexity. Note that 2-4[04] also occurs three times, in α_3, α_5 and α_6, and in melodic form between notes B and D♯. However, Example 15



EXAMPLE 15(a). *Plot in the FQS of phrases α (black) and δ (gray). Phrase centroids appear with squares, RSs with large circles, and pcsets with small circles.*



EXAMPLE 15(b). *Rendition of phrases β (black) and ϵ (gray). Similarly to Example 15(a), centroids appear with squares, RSs with large circles, and pcsets with small circles.*



EXAMPLE 15(c). *Depiction of two more phrases from Schoenberg's Op. 19, No. 1 in the FQS, namely phrases γ (black) and ζ (gray). Centroids appear with squares, RSs with large circles, and pcsets with small circles.*

shows that α_5 and α_6 have a strong influence in the quale of phrase α since their chromatic component attracts its centroid, G_α, towards the $\hat{f}_1$ and $\hat{f}_2$ regions. This fact favors [02] as the RS_α by the fact that [04] would

imply a more preponderant role for octatonicity and triadicity in the phrase.

The RS_{β} is the last pcset in β, i.e., the pentachord {0,3,5,9,A} ↦ 5-29[01368], or β_{19}. This pcset is somehow reminiscent of phrase α since (1) {3,5} and {0,A} strongly represent interval-class 2 in β_{19}, and (2) one instance of interval-class 2 happens with D♯ and F (in phrase α, these notes appear melodically and in pcset α_6).[65] Furthermore, β_{19} also stands as a coherent representative of β since its subset {0,3,9,A} ↦ 4-13[0136] is heard also in β_{12} and β_{16}.[66]

Tetrachord γ_1, {2,3,6,9} ↦ 4-18[0147], is the RS_{γ}. The appearance of [0147] as an RS is predictable. Kramer identifies it as one of the central motifs driving the horizontal and vertical development of the entire piece.[67] For instance, notes A, B♭, F♯ and D♯ in the right hand of m. 2 or notes F, E, D♭ and B♭, opening this same measure in the left hand, denote its horizontal importance. Finally, the RSs of δ, ϵ, and ζ are pcsets δ_3, ϵ_3, and ζ_8, or {2,6} ↦ 2-4[04], {1,3,5,9,A} ↦ 5-30[01468], and {3,4,6,8,A} ↦ 5-24[01357], respectively. Their proximity to the RS_{γ} is evident since they share interval properties with 4-18[0147]. Moreover, identifying 2-4[04] as an RS is noteworthy because it assumes great preponderance in No. 1 and the remaining pieces of Schoenberg's Op. 19, especially in No. 2.

4.2 QUALIA MOTION AND ANALYTICAL REMARKS ON THE PROGRESSION OF REFERENCE SONORITIES

Example 16(a) shows the qualia motion of the piece resulting from RS_{α} to RS_{ζ}. It highlights an anticlockwise direction in the FQS that starts and ends in the dyadicity region. In detail, the regions linked with each phrase are as follows: α ⟶ dyadicity, β ⟶ diatonicity, γ ⟶ triadicity, δ ⟶ octatonicity, ϵ ⟶ octatonicity, and γ ⟶ dyadicity.

Moreover, Example 16(b) schematically shows two crucial relationships. Dyadicity is the first and final quality in the trajectory, yet it is absent in all remaining phrases. Such a pattern establishes a distant relation in No. 1's structure. However, it promotes its vision as an anchor that settles the qualia motion's beginning and end as if it were a familiar sonority. On the other hand, the fact that octatonicity repeats in two consecutive phrases as the piece approaches its end—until there is no repeated quality—generates a slowing of the quale.[68] Thus, the sparser harmonic rhythm perfectly serves the piece's closure.

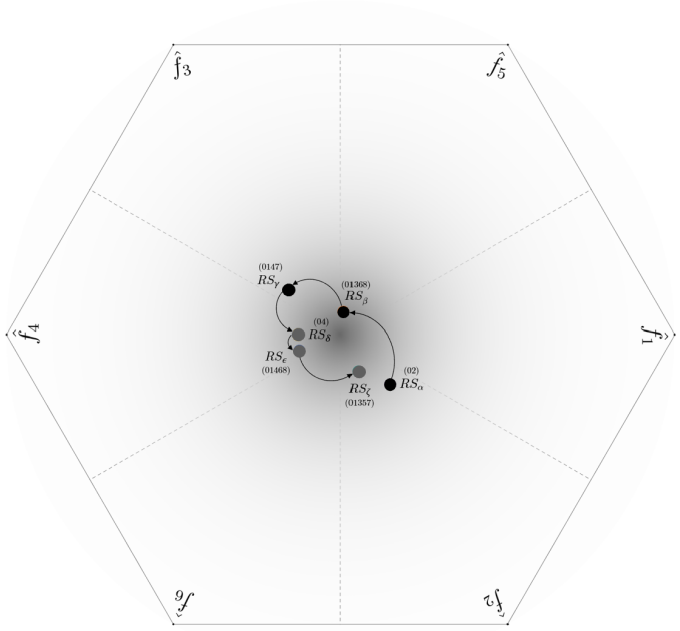
Another relevant aspect is that chromaticity and whole-tone quality are absent from the piece's qualia motion. One hint to this fact is the predominance of set class (0147). This set class produces $\left|\hat{f}_1\right| = 1$; $\left|\hat{f}_2\right| = 1{,}732$; $\left|\hat{f}_3\right| =$

[65]The note F is peculiar in β since it appears only two times in the phrase, precisely in β_1 and β_{19}. Thus, it connects the beginning and ending of β's harmonic gesture.

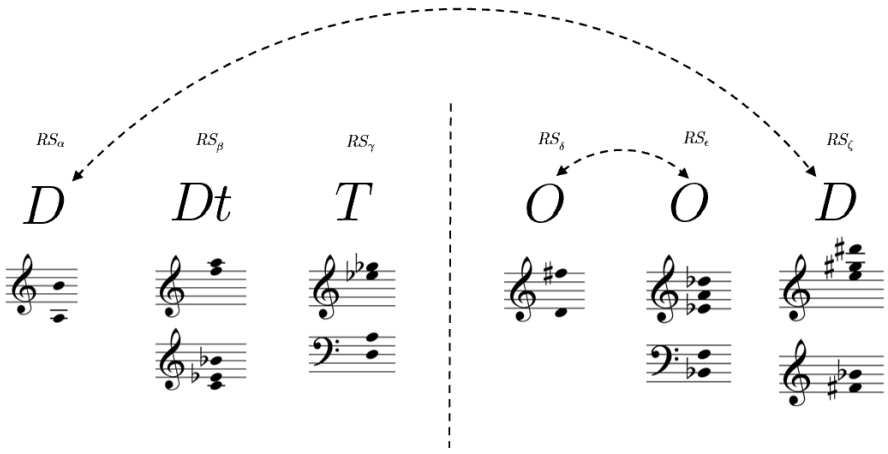
[66]There is also pcset β_5, with pitch-classes {0,6,7} and set class 3-5[016], that being also a subset of 5-29[01368] and 4-13[0136], attracts its centroid G_{β} into the region of [01368].

[67]Kramer (1988, 170–83).

[68]The expression "qualia slowing down" alludes to harmonic rhythm, or, more precisely, qualia rhythm, which slows down by the repetition of octatonicity.

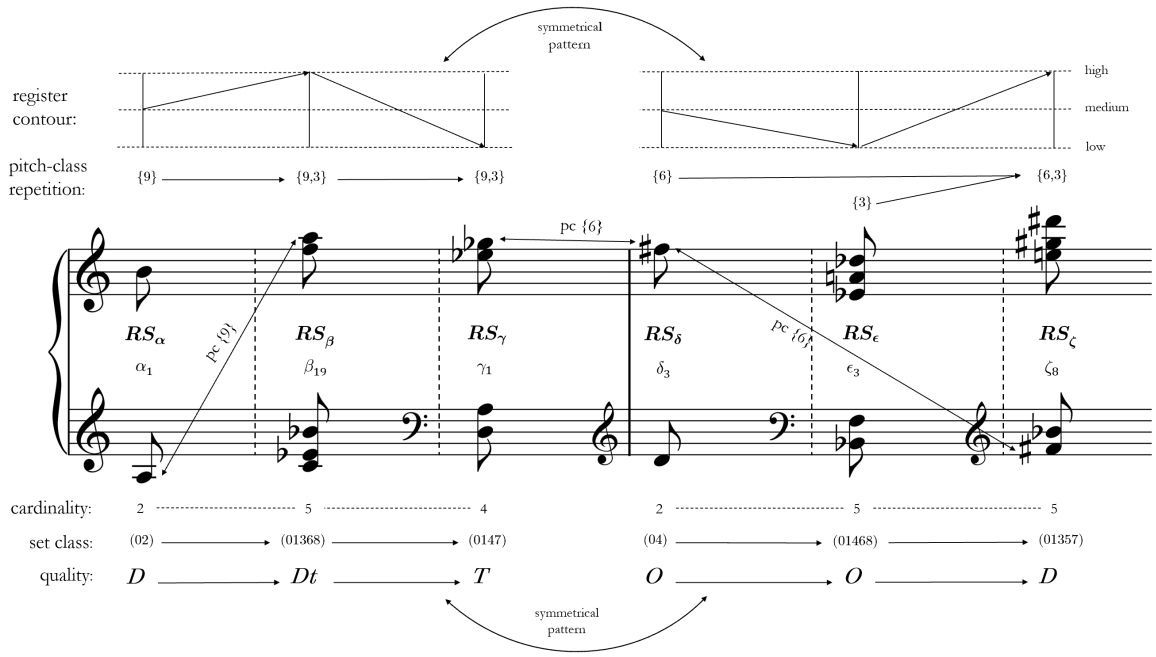


EXAMPLE 16(a). *The trajectory of RSs in Schoenberg's Op.19/1 evidencing an anticlockwise circular motion.*



EXAMPLE 16(b). *The progression of RSs in score and relationships between its harmonic qualities. D, Dt, T and O stand for diadicity, diatonicity, triadicity and octatonicity, respectively.*

2; $\left|\hat{f}_4\right| = 2{,}646$; $\left|\hat{f}_5\right| = 1$, and $\left|\hat{f}_6\right| = 0$. Its coefficients $\hat{f}_4$ and $\hat{f}_2$ are the most preponderant. On the opposite side, $\hat{f}_6$ and $\hat{f}_1$, referring precisely to whole-tone and chromaticity, have low values. These facts elucidate the results obtained and state the preponderance of 4-18[0147] in the piece since dyadicity and octatonicity are the only repeated qualities and by the inexistence of chromaticity and whole-tone qualities in the trajectory. Furthermore, it is worth examining whether the identified RSs are manifestations of the now-identified trajectory, namely through their harmonic qualities. Each RS has interval properties aligned with the region it occupies, except for the cases of RS_α and RS_δ, whose low cardinality hinders a clear association between them and harmonic qualities. The RS_β emerges as diatonic since it represents a subset of the C minor scale. The RS_γ, which favors both minor and major thirds over other intervals—⟨1,0,2,1,1,1⟩—stems in the triadicity region. The RS_ϵ appears as octatonic since four of its five notes form an octatonic subset, i.e., {1,3,9,A}. Finally,



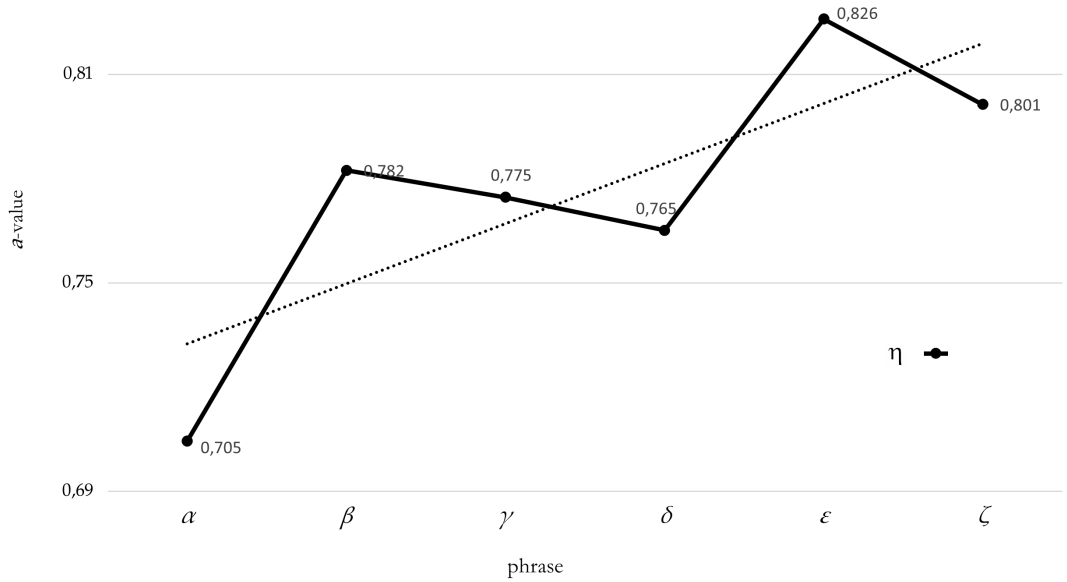
EXAMPLE 17. *This Example outlines several aspects that support a symmetrical and bipartite vision of the progression of RSs. From top to bottom, one can analyse the graphical representation of the progression's contour in register, the repetition of pitch-classes* $\{3, 6, 9\}$*, the RSs' cardinality, set classes and their harmonic quality in FQS. D, Dt, T and O stand for diadicity, diatonicity, triadicity and octatonicity, respectively.*

RS_ζ positions itself slightly in the dyadicity region since three of its five notes—{3,4,A}—are a tritone and a minor second, relating this pcset to $\hat{f}_2$.

On the other hand, it is challenging to link RS_α and RS_δ to the dyadicity and octatonicity qualities due to their low cardinality. However, observing the positioning of G_α and G_δ, one can verify that both occupy the same regions of RS_α and RS_δ, thus forming a similar trajectory to the one in Example 16(a). As such, it is reasonable to interpret α, δ, and their RSs as elements of dyadicity and octatonicity.

The set class domain reinforces a bipartite vision of the piece. A subtle similarity exists between the first three RSs and the final three, establishing a relationship between their set classes. Specifically, RS_α corresponds to RS_δ, RS_β corresponds to RS_ϵ, and RS_γ corresponds to RS_ζ. In the set class domain, 2-2[02] corresponds to 2-4[04], 5-29[01368] corresponds to 5-30[01468], and 4-18[0147] corresponds to 5-24[01357]. This similarity is evident not only in the cardinalities of the RSs but also in the resemblance of their interval properties. This relationship reinforces the two-part division of the work and highlights the consequential effect on the harmonic material of δ, ϵ, and ζ. The idea of an "antecedent" and "consequent" resonates whenever we analyze the progression, and Example 17 summarizes it. For example, there is a symmetry in register between $RS_{\alpha\to\gamma}$ and $RS_{\delta\to\zeta}$. The first group presents a contour in register such that $RS_\alpha \mapsto mid$, $RS_\beta \mapsto high$, and $RS_\gamma \mapsto low$, whereas the second is an inversion, i.e., $RS_\delta \mapsto mid$, $RS_\epsilon \mapsto low$, and $RS_\zeta \mapsto high$. Such inverted contour reinforces the sensation of consequence in $RS_{\delta\to\zeta}$, promoting a sense of inevitability in the progression.

Furthermore, the repetition of certain pitch classes also contributes to this consequential effect. Pitch-class 9 is present in $RS_{\alpha\to\gamma}$, and pitch-class 3 is pervasive in RS_β and RS_γ. On the other hand, pitch-class 6 and pitch-



EXAMPLE 18. *Qualia ambiguity in Schoenberg's Op.19, No.1. The solid line stands for the computed values of ambiguity (η) and the dashed line shows the linear tendency of η, showing a positive slope and increasingly ambiguous qualia in the piece.*

class 3 have a related pattern in $RS_{\delta\rightarrow\zeta}$. Moreover, such an effect is also visible through the inverted position that pitch-classes 9 and 6 occupy in their respective pcsets. For instance, pitch-class 9 is the lowest note in RS_{α}, becoming its highest in RS_{β}, and in an inverted behaviour, pitch-class 6 in RS_{δ} is the highest note of the pcset, and in RS_{ζ}, it becomes the lowest.

4.3 AMBIGUITY IN OP. 19, NO. 1'S HARMONIC QUALIA: EMERGING LINEARITY AND COUNTERPOINT

Qualia ambiguity in Op. 19, No. 1 reinforces the bipartite structure and provides a new parameter to support Kramer's nondirected linearity. Example 18 shows that the η values are moderate to high in all phrases, ranging between .71 and .83, and the average ambiguity value for all six phrases is $\overline{\eta} = .78$. Globally, these values suggest ambiguous music in the harmonic qualia domain, a fact already expected given the compositional period in question and the most structural pcsets in the piece. For example, η_{α} is the lowest value, but its harmonic profile is not definitively associated with any quality, as would be expected in a low η hypothesis. Even so, pcsets α_4 and α_8 (shown in Ex. 15), associated with triadicity and whole-tone quality, make this phrase the least ambiguous.

Nonetheless, Example 18 reveals a clear tendency of pcset concentration around the center that progressively increases as the piece unfolds phrase by phrase, meaning, by definition, an increasingly ambiguous quale. The positive slope of the regression line clearly expresses this tendency. Thus, successive advances in phrasal unfolding tend to correspond to successive increases in ambiguity. Moreover, listening to the piece seems to suggest a *crescendo* of qualia ambiguity since its unfolding induces the listener to the sensation of an increasingly mysterious and ambiguous harmonic character. It is also curious that this trend arises in an

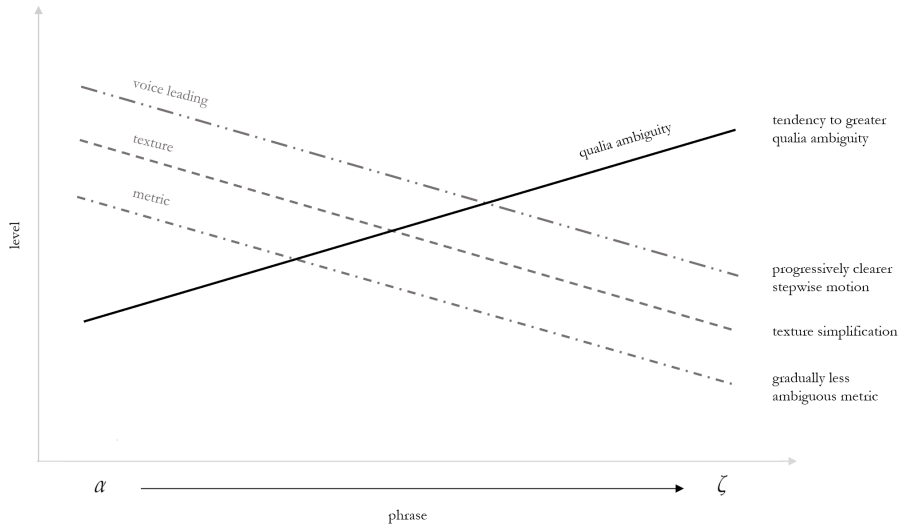
increasing and not decreasing direction since Schoenberg's Op. 19, No. 1 epitomizes the broader trajectory of the emancipation of dissonance and the crisis of tonality in the compositional context of the late nineteenth and early twentieth centuries.

The moments of increasing and decreasing ambiguity form a symmetrical outline reminiscent of the progression of RSs in Example 17 and its bipartite vision. Right from the outset, the work exhibits a significant surge in ambiguity from α to β, and a strikingly similar surge recurs from δ to ϵ, thereby establishing δ as a symmetry axis within the structure of the composition. Furthermore, when analysing η's ascending and descending profile in the second part, it is impossible not to notice the symmetry between ambiguity and the melodic contour of the progression of RSs as a noteworthy contrapuntal aspect. In $\alpha \rightarrow \gamma$, the ambiguity curve outlines a retrograded contour in relation to the melodic contour, namely $\eta_\alpha = .705 \mapsto \textit{low}$, $\eta_\beta = .782 \mapsto \textit{high}$, and $\eta_\gamma = .775 \mapsto$ mid, whereas in $\delta \rightarrow \zeta$, broadly speaking, we have $\eta_\delta = .765 \mapsto$ low, $\eta_\epsilon = .826 \mapsto \textit{high}$, and $\eta_\zeta = .801 \mapsto \textit{mid}$, i.e., an inverted and retrograded motion.

However, despite the increasing tendency in ambiguity, Example 18 exhibits three moments of decrease in the η curve: in $\beta \rightarrow \gamma$, $\gamma \rightarrow \delta$, and $\epsilon \rightarrow \zeta$. In δ, pcset δ_8—visible in Example 15—greatly contributes to a decrease in ambiguity due to its proximity to the whole-tone anchor. Similarly, ζ has a decrease in ambiguity and shares textural properties with δ. In both, the recourse to a pcset with a long duration is distinctive, emerging above or below a melody that slightly directs quale to distinct places without losing the stability that the longer pcset embodies. The similar texture in phrases δ and ζ helps us understand their increase in η due to the effect of the longer sonority, that stabilizes the two phrases in the harmonic qualia domain.

Moreover, Example 18 reveals ambiguity as an element of nondirected linearity since it configures a tendency that propels the piece forward along a path without an evident and predictable destination. However, we notice that the qualia ambiguity of Op. 19, No. 1 assumes a contrapuntal character with respect to other linear aspects. Kramer maintains that voice leading and metric motions gradually become less ambiguous throughout the piece. The former reveals the emergence of stepwise motion in the melodic web, while the latter becomes more explicit throughout the piece, with an increasing sense of metrical downbeats and upbeats. In parallel, Kramer also points to a gradual simplification in texture, thus accompanying the idea of progressively less ambiguous linear elements.

In this way, it is perceptible that qualia ambiguity in Op. 19, No. 1, although a linear element, moves in opposite direction from voice leading, metric, and texture motions—Example 19 diagrams this idea. Therefore, we notice that the harmonic qualia appear in counterpoint, evidencing a contrary motion that gives them independence instead of a parallel movement that would make all linear elements dependent among themselves. From our point of view, such contrapuntal linearities reinforce the sensation of musical discourse in progressive and nondirected movement since they create a complex network of independent musical structures, although interdependent. This intricate and contrapuntal web marks the musical discourse of Schoenberg's Op. 19, No. 1.



EXAMPLE 18. *Non-directed linearities in Op.19/1 in contrary motion. The counterpoint emerges between voice leading, texture, metric, and harmonic qualia: parallel motion in the first three and contrary motion between them and harmonic qualia. The y-axis stands for a non-quantifiable measure of ambiguity.*

CONCLUSION

We argue that the DFT of pcsets enables the conceptualization of a multidimensional space where chromaticity, dyadicity, triadicity, octatonicity, diatonicity and whole-tone quality coexist. To account for the interaction between coefficients, we propose the FQS—a space confined to the area of a hexagon, where the positioning of set classes depends on their magnitude saliency to the preceding qualities. Ordering the DFT coefficients in the FQS, such as $\hat{f}_1$, $\hat{f}_5$, $\hat{f}_3$, $\hat{f}_4$, $\hat{f}_6$, and $\hat{f}_2$, results in a Pearson correlation coefficient of $r = .729$, i.e., a positive correlation that preserves 73% of the set classes' Euclidean distances in relation to a six-dimensional space. The distortion introduced by our space, resulting from its convex structure, allows the definition of a metric for qualia ambiguity. Applying it to a large musical corpus and different compositional periods would help understand its value for diachronic analysis.

Furthermore, we advance a mathematical system inspired by Mailman and grounded on the FQS to identify a set class from a musical phrase. This set class is an RS, i.e., the best representation of the phrase's harmonic qualities. When one arranges the RSs in a piece, one obtains a reduction of its qualia motion. At this point, it is essential to consider whether an RS perceptually represents the qualia content of a phrase in free atonality. Unlike the music of Debussy, in which the listener identifies passages associated with a harmonic quality, it is difficult to argue that if the RS of an atonal phrase is octatonic, the listener will undoubtedly perceive it as octatonic. However, our approach to selecting an RS democratically considers all the phrase's pcsets and chooses the one that best summarizes it, even if this distinction is audibly and perceptually tricky.

Our analytical considerations of Schoenberg's Op. 19, No. 1, concentrate on harmonic elements, leaving aside other essential ones such as rhythm, texture, dynamics, or timbre, with the vicissitudes that this fact entails. On the one hand, it allows us to look in detail at harmonic qualia in a repertoire that usually does not focus on it, but on the other, it is undeniable that a robust and complete analysis of the piece needs to look at the interaction between all the above elements.

Nevertheless, the FQS reinforces the structural role of set class 4-18[0147]. Not only does it emerge as an RS, but it also is the "gravitational center" around which the other RSs—2-2[02], 5-29[01368], 2-4[04], 5-30[01468], and 5-24[01357]—orbit. Moreover, the progression of RSs suggests dividing the work into two significant time spans: the first between mm. 1 and 6 and the second between mm. 7 and 17. The progression's multi-level symmetry supports such a division, which can be seen either through the cardinalities and set classes of the RSs or through the inverted contour in the register identifiable in Example 17. These facts, coupled with the progression's circular trajectory in FQS, in a back-and-forth pattern visible in Example 16(a), highlight the bipartite perspective of the piece and give the second part of the progression the "consequent" character that presents an answer to its respective question.

Finally, the analysis of the qualia ambiguity presents globally high values and points to an increasing tendency across the piece. These results allow us to note two essential ideas: (1) They reveal ambiguity as an element of Kramer's nondirected linearity that propels the music forward and keeps it moving without generating a concrete expectation of resolution, and (2) they evidence contrary motion between different elements of nondirected linearity.[69] Voice leading, texture, and meter show gradual simplification, while ambiguity tends to complexification. Such counterpoint affects Op. 19, No. 1's fluidity and preserves the independence and singularity of each structure.

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[69]Even if we have obtained dyadicity as the piece's first and final harmonic quality, it is difficult to claim that its end guarantees the listener a clear sense of resolution and complete rest, as if dyadicity played a "tonic" syntactic role. The unfolding of the piece suggests motion and progression but does not create enough expectation capable of causing the listener to predict any concrete resolution.

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